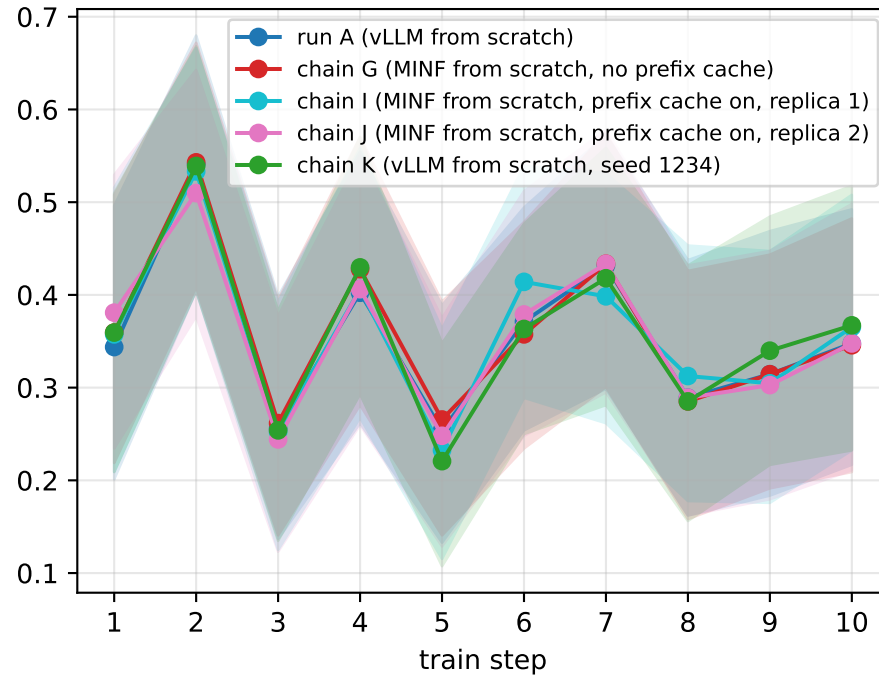
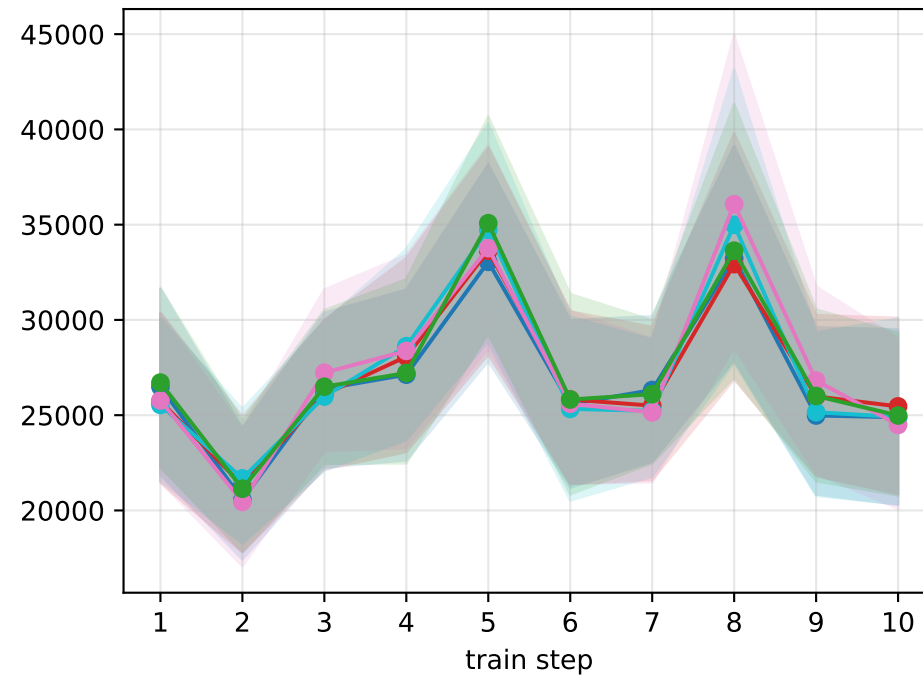


Per-step overview, steps 1-10 (bands = 95% prompt-cluster bootstrap CI of the step mean)

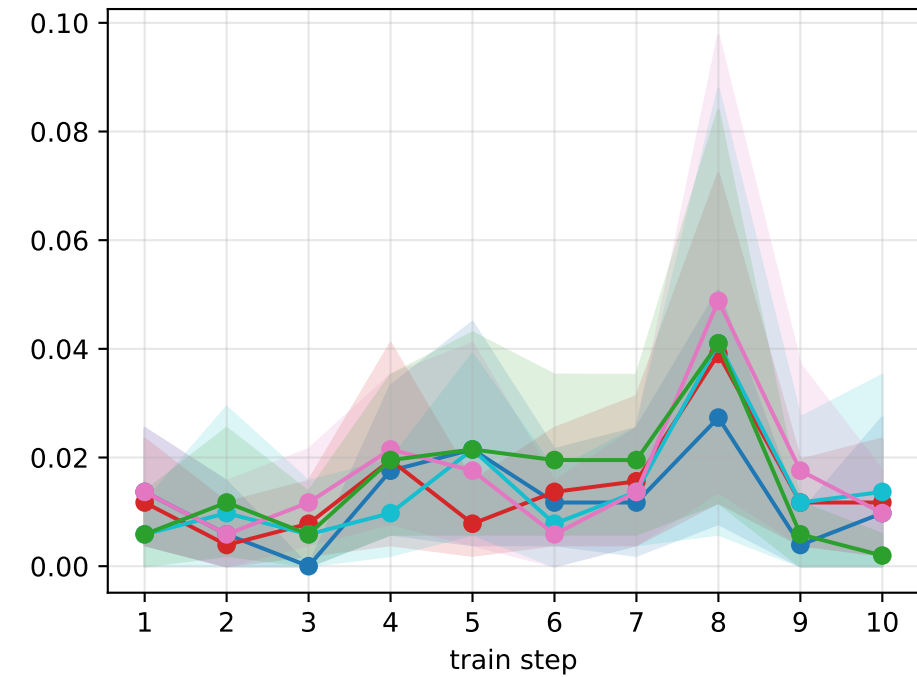
mean reward (share resolved)



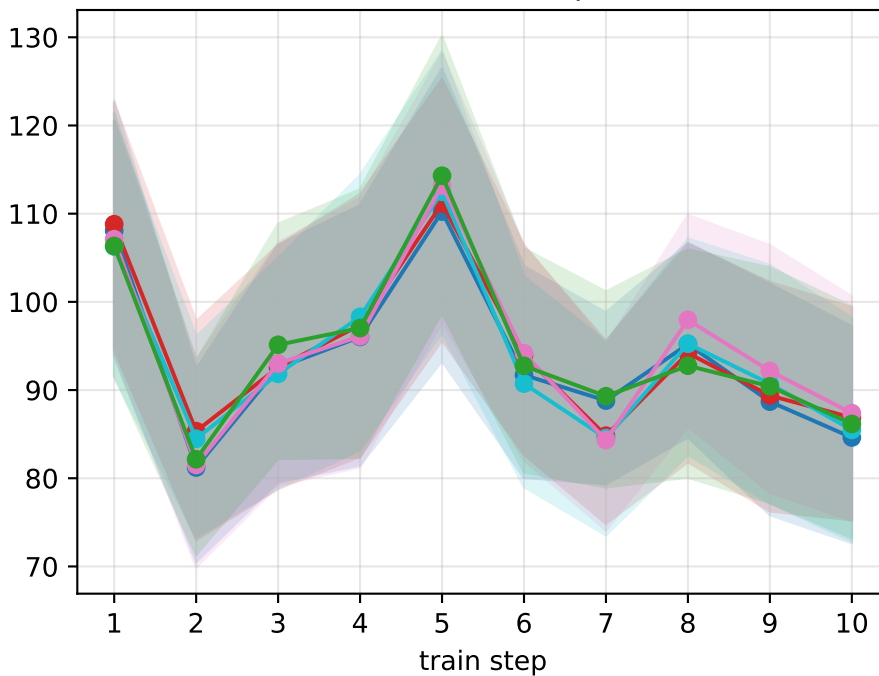
mean generated tokens per rollout



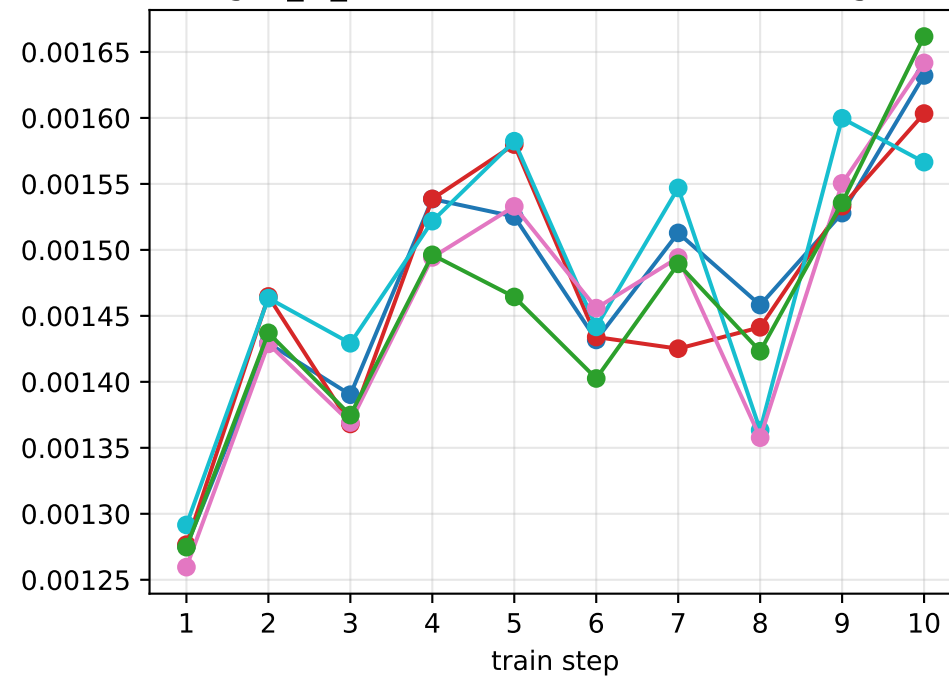
context-window truncation rate



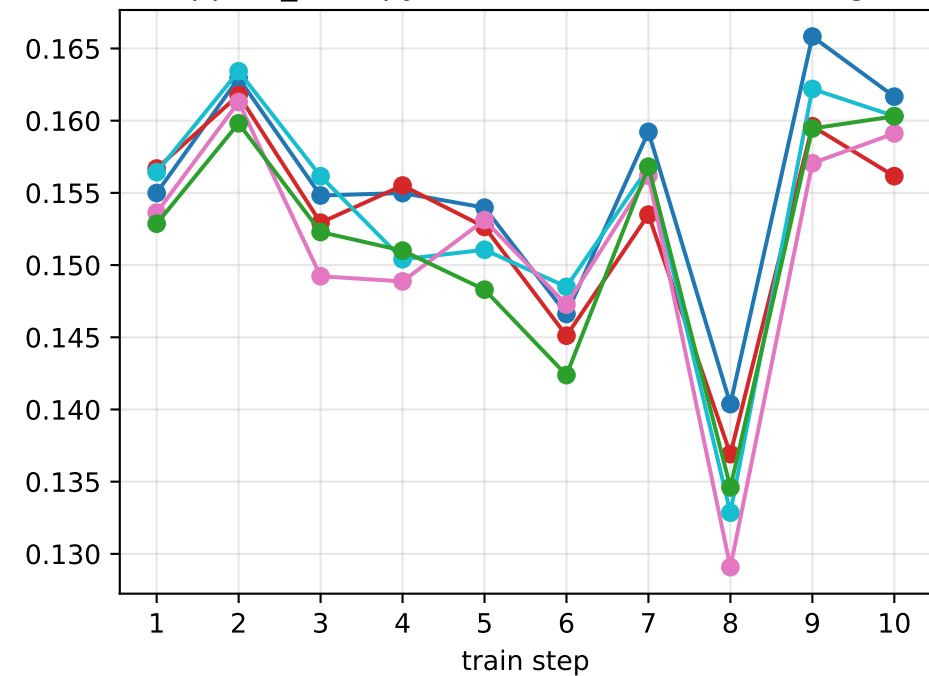
mean assistant turns per rollout



gen_kl_error (trainer, from controller log)

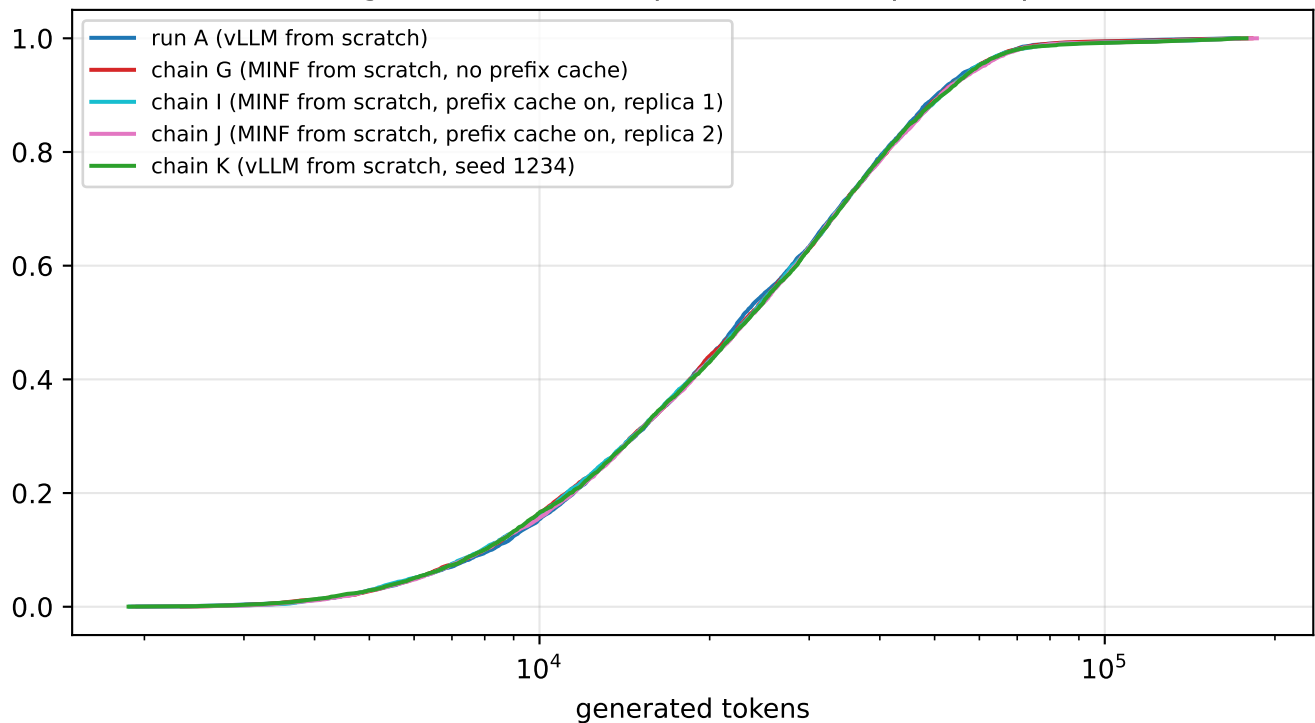


approx_entropy (trainer, from controller log)

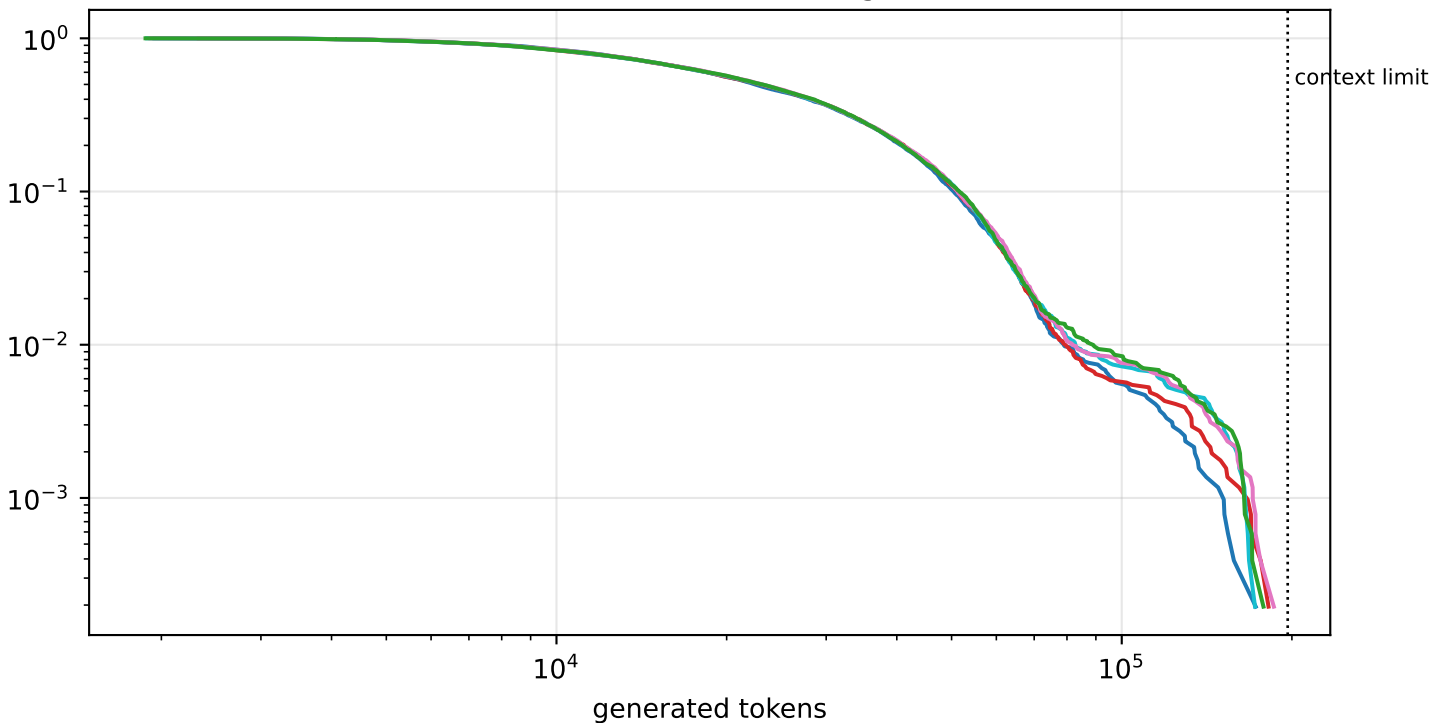


Generation length distribution including the tail

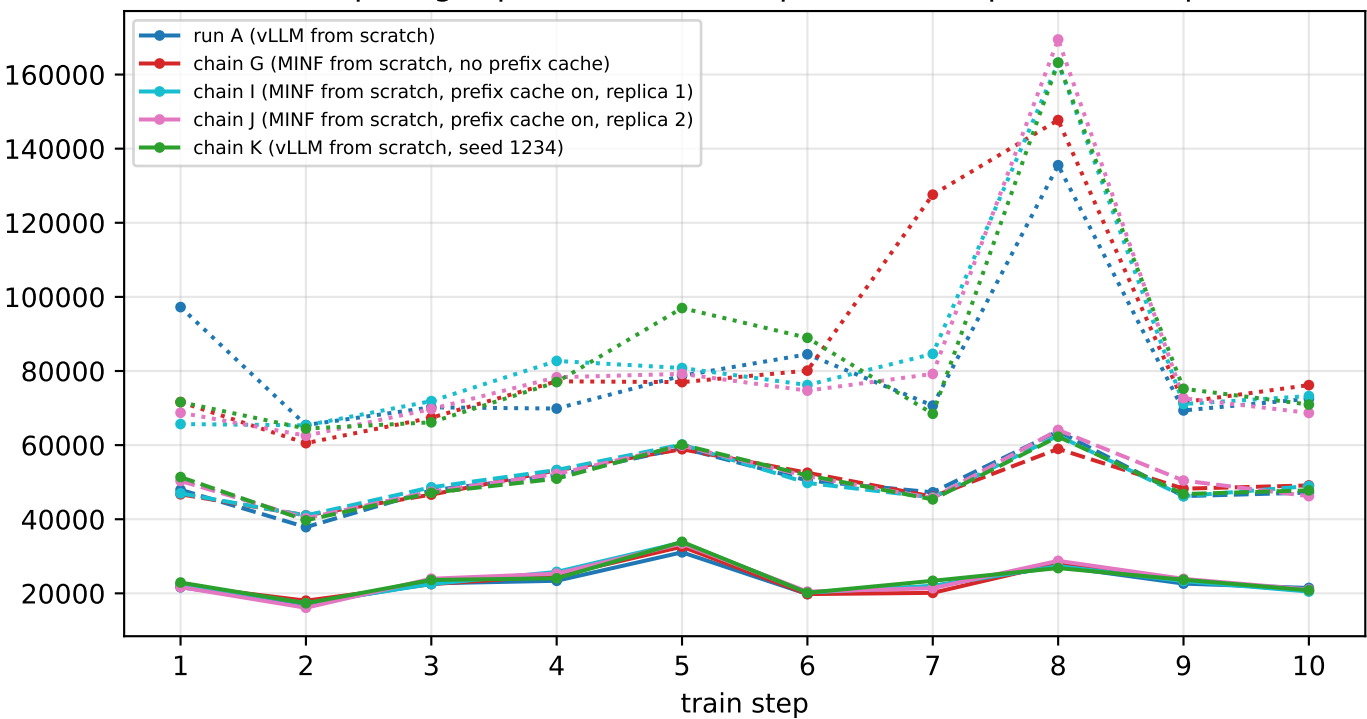
CDF of generated tokens per rollout (steps 1-10 pooled)



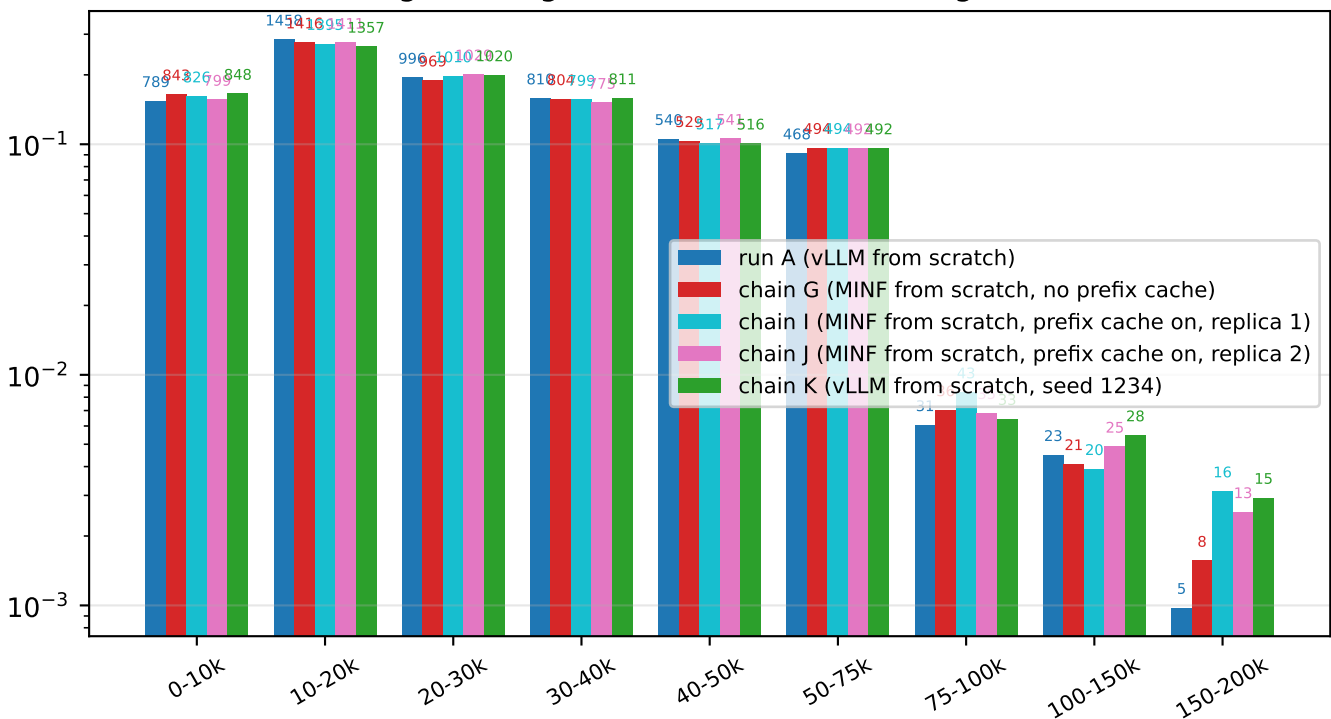
Tail: survival function P(length > x)



Per-step length percentiles: solid p50, dashed p90, dotted p99

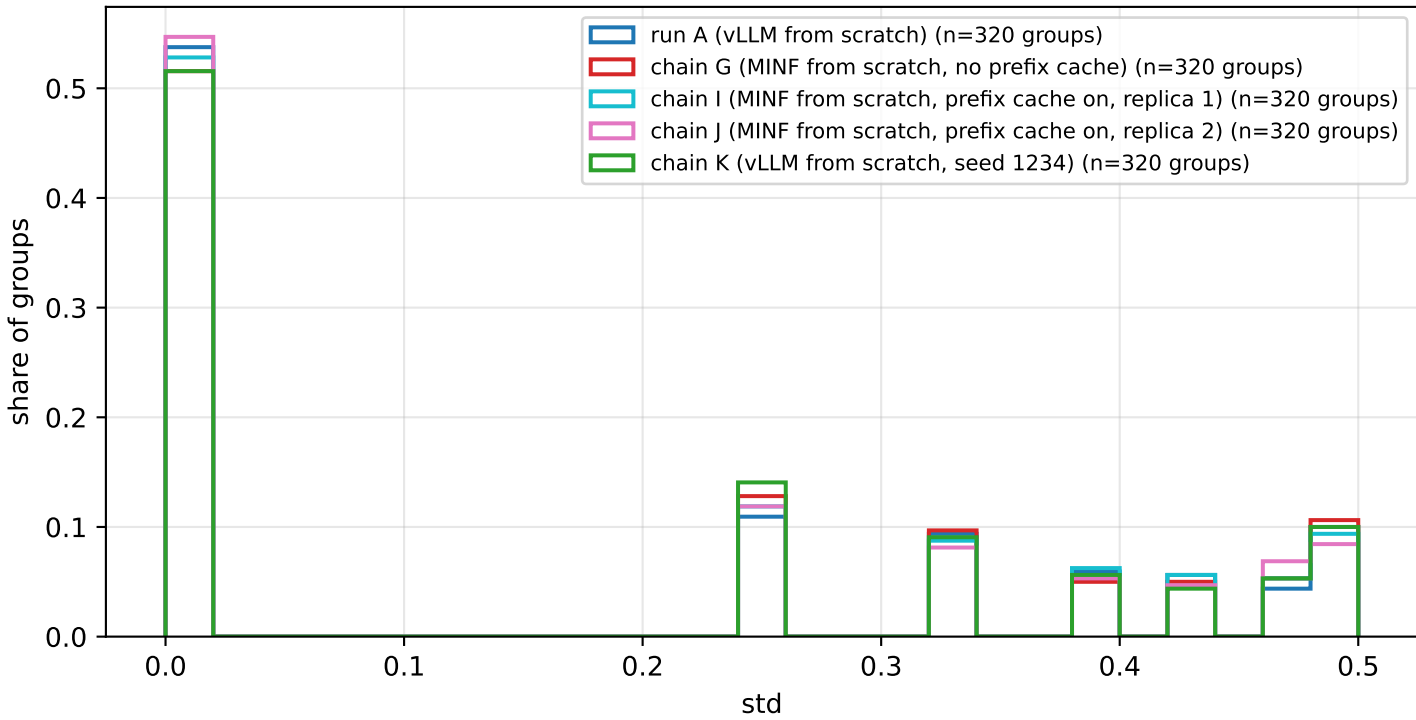


Length histogram (share of rollouts, log scale)

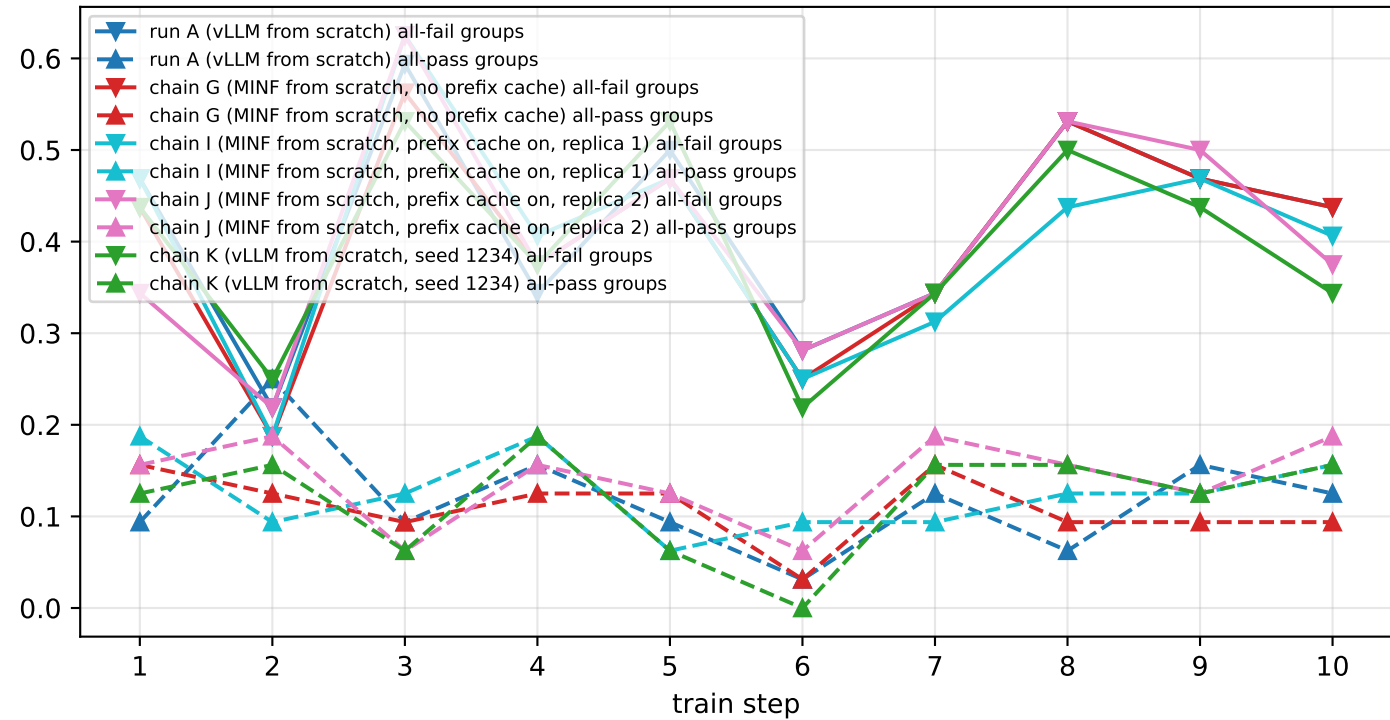


Reward and GRPO group structure

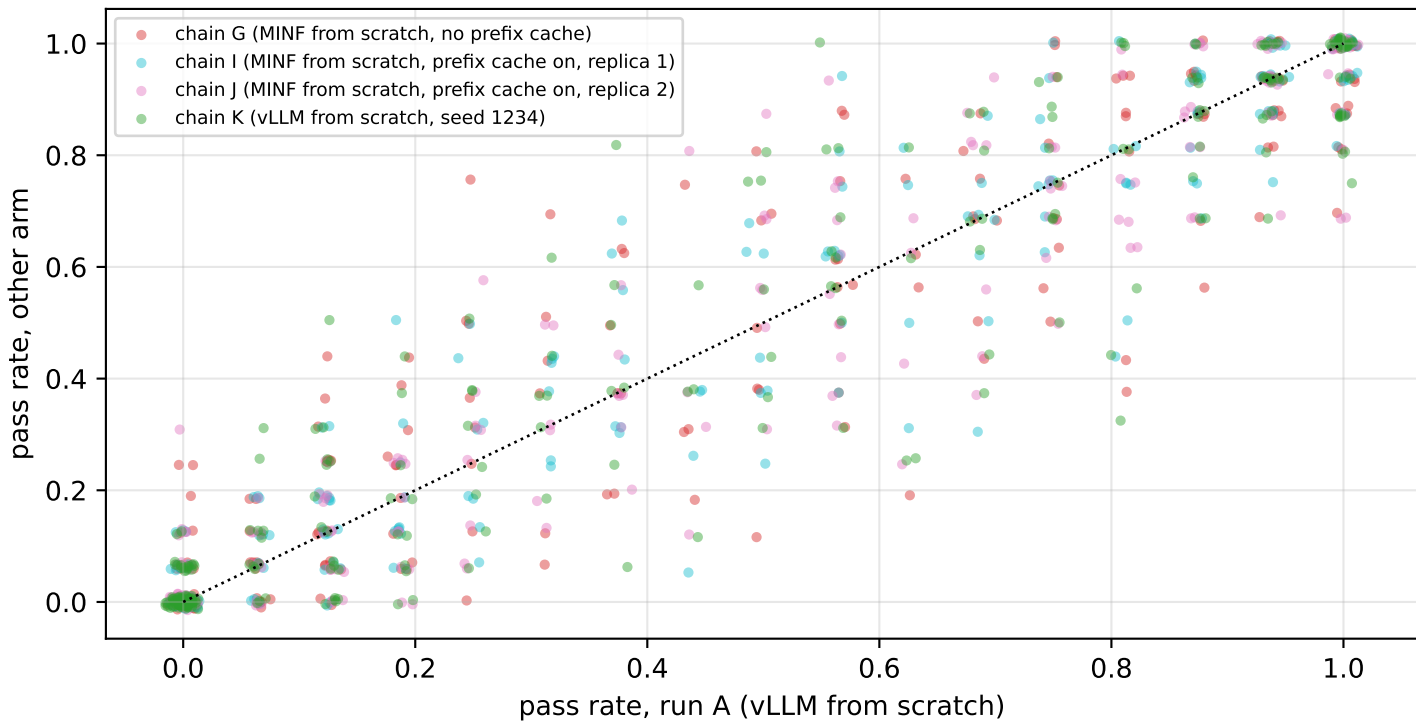
Within-group reward std (16 rollouts per group, ddof=0)



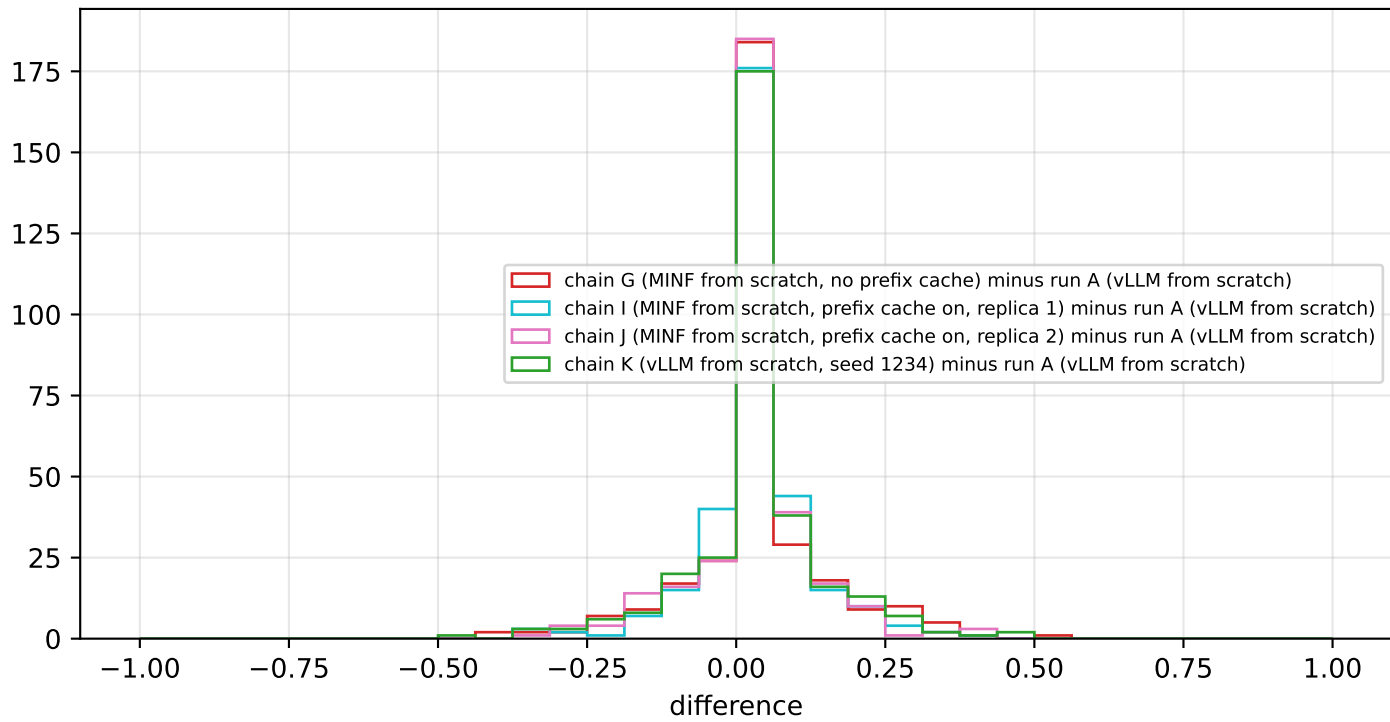
Share of degenerate groups (zero advantage signal)



Per-prompt pass rate vs run A (vLLM from scratch) (matched prompt-steps)

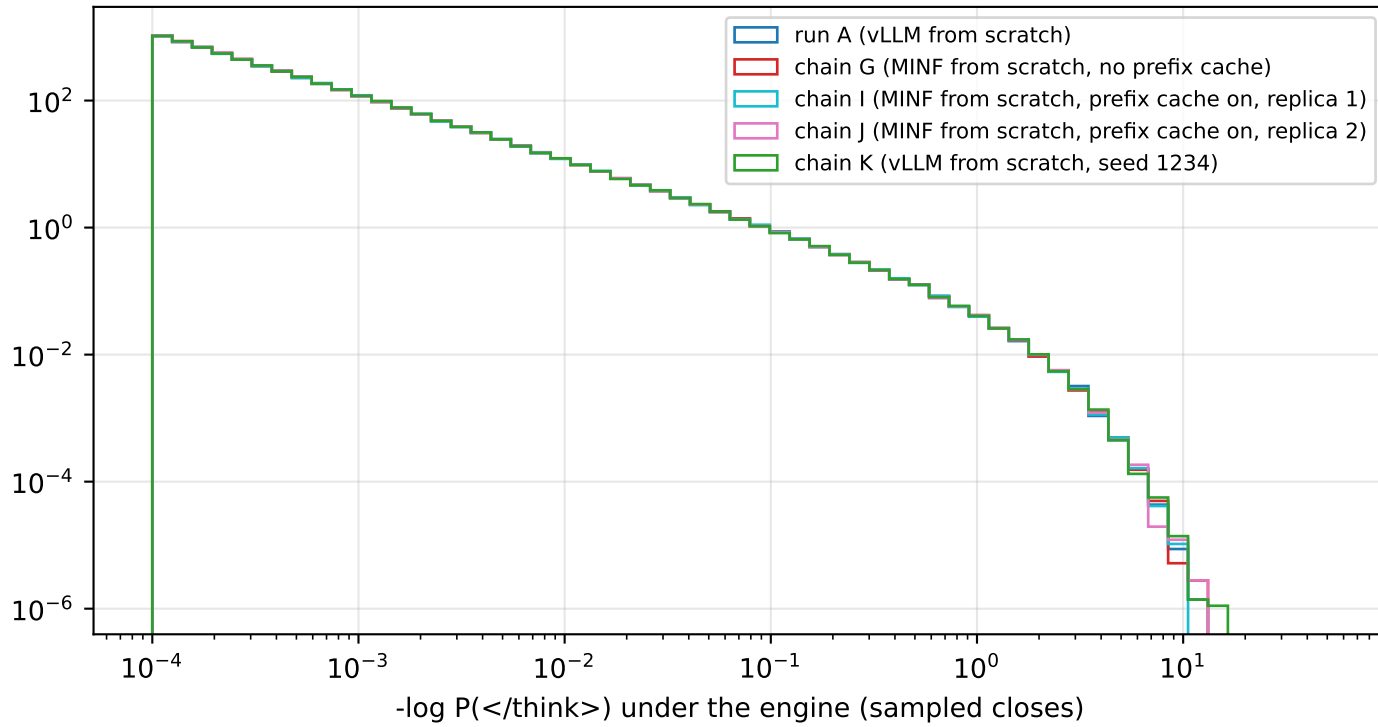


Paired pass-rate difference per prompt, other arm minus run A (vLLM from scratch)

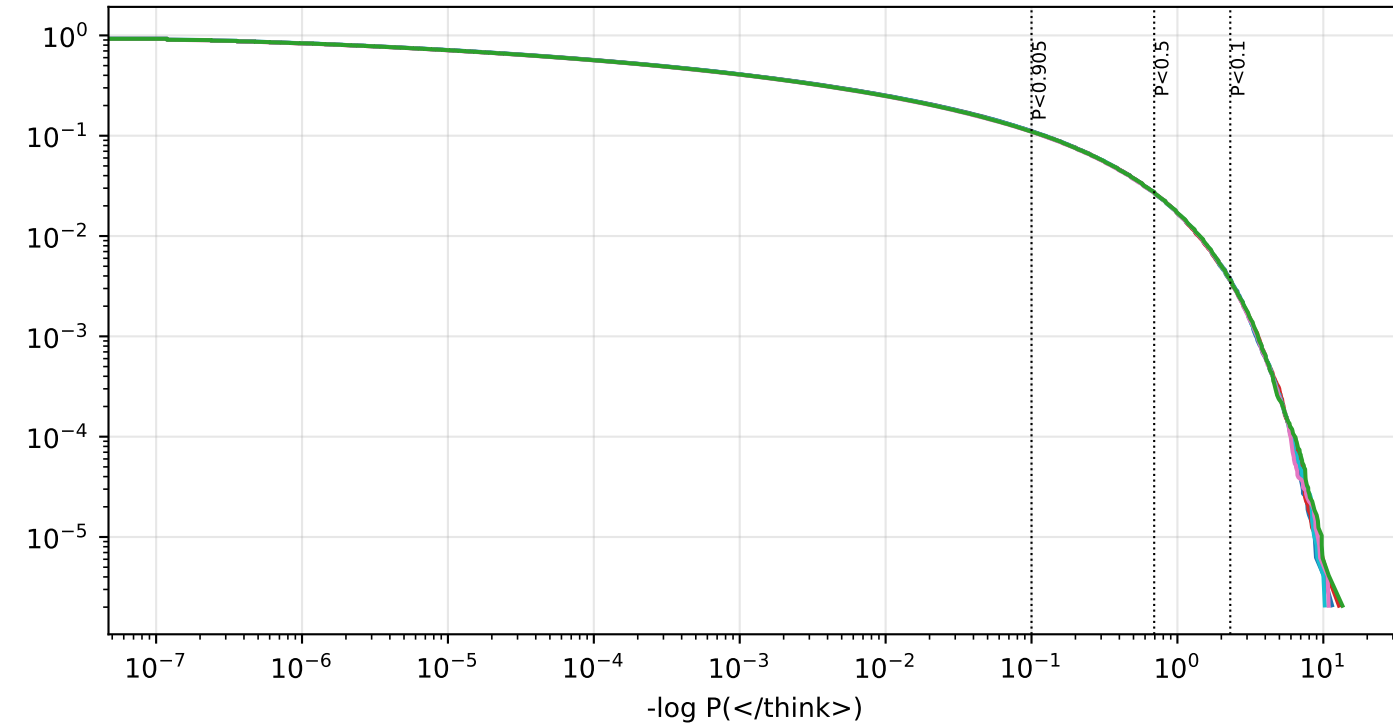


End-of-thinking token: engine probability of the sampled </think>

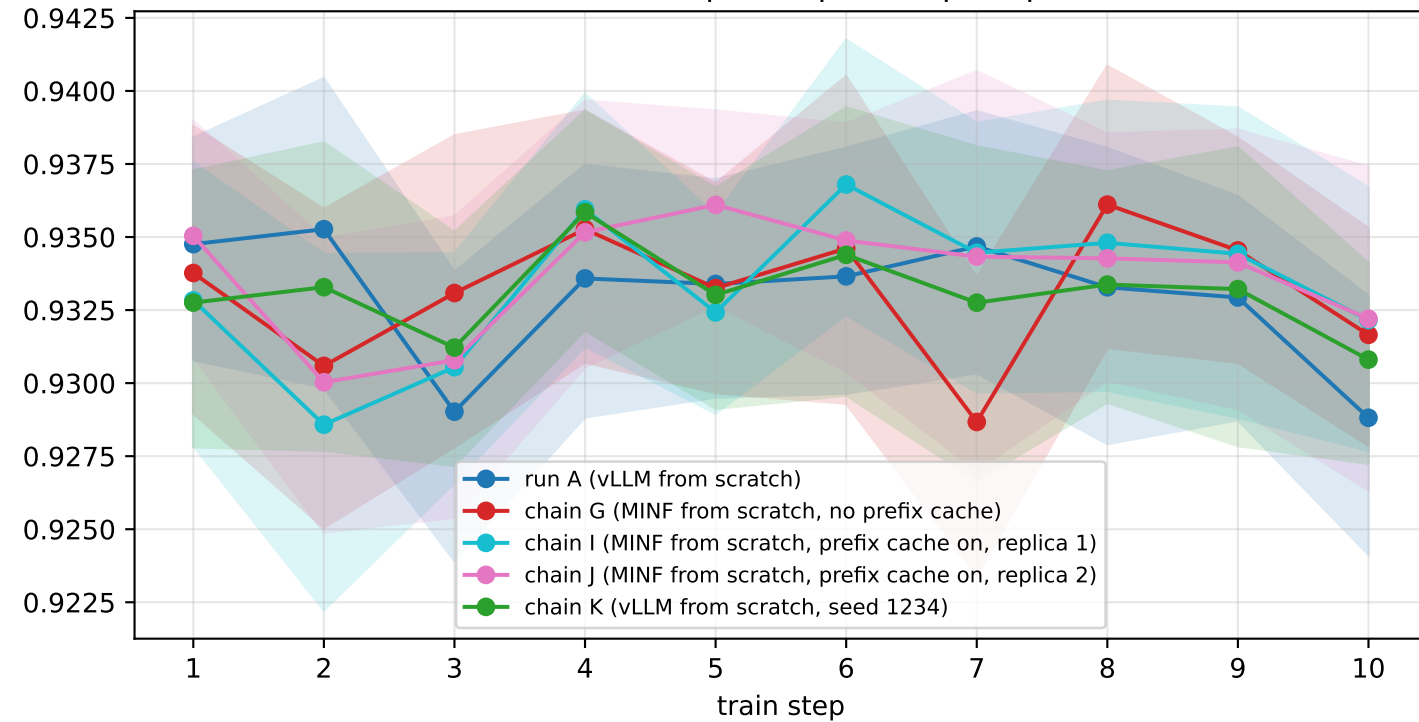
Distribution of the engine's </think> logprob



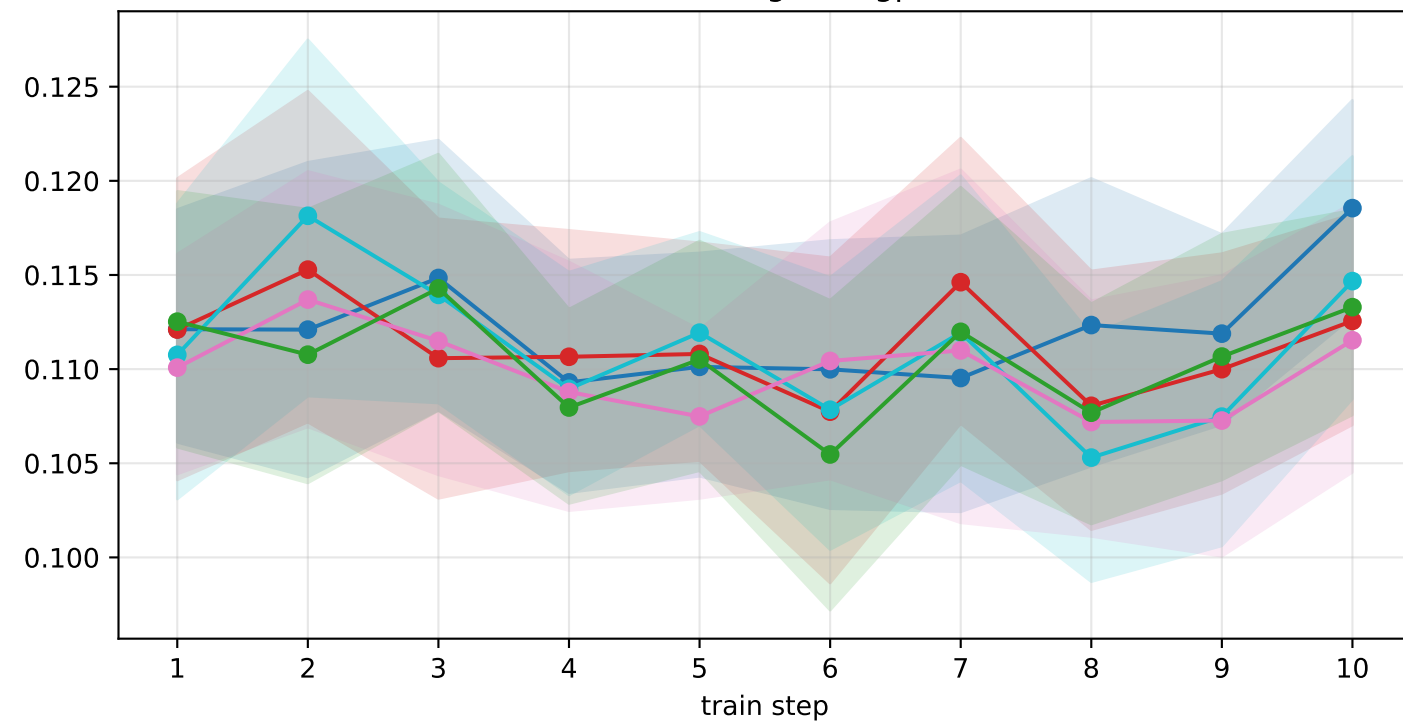
Tail: share of closes with $-\log P > x$



Geo-mean $P(</think>)$ per step (95% prompt-cluster CI)

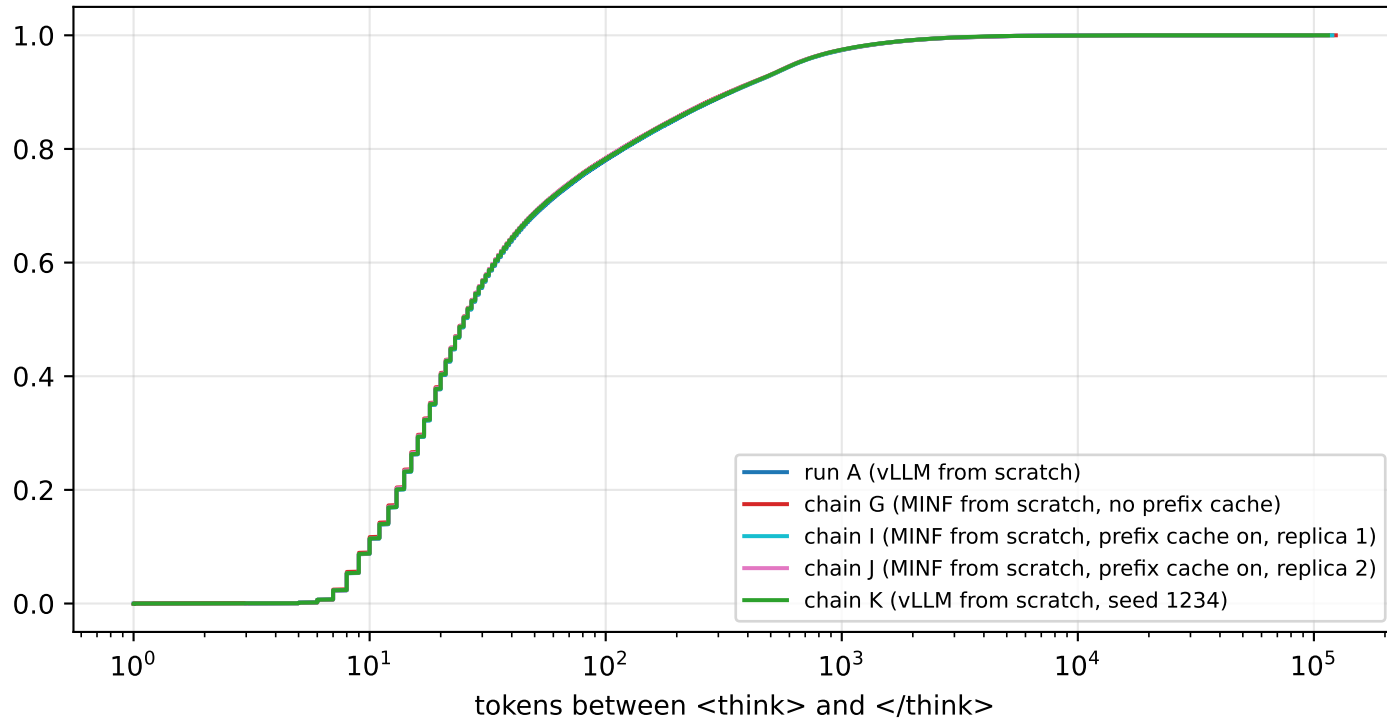


Low-confidence close share (engine logprob < -0.1 , $P < 0.905$)

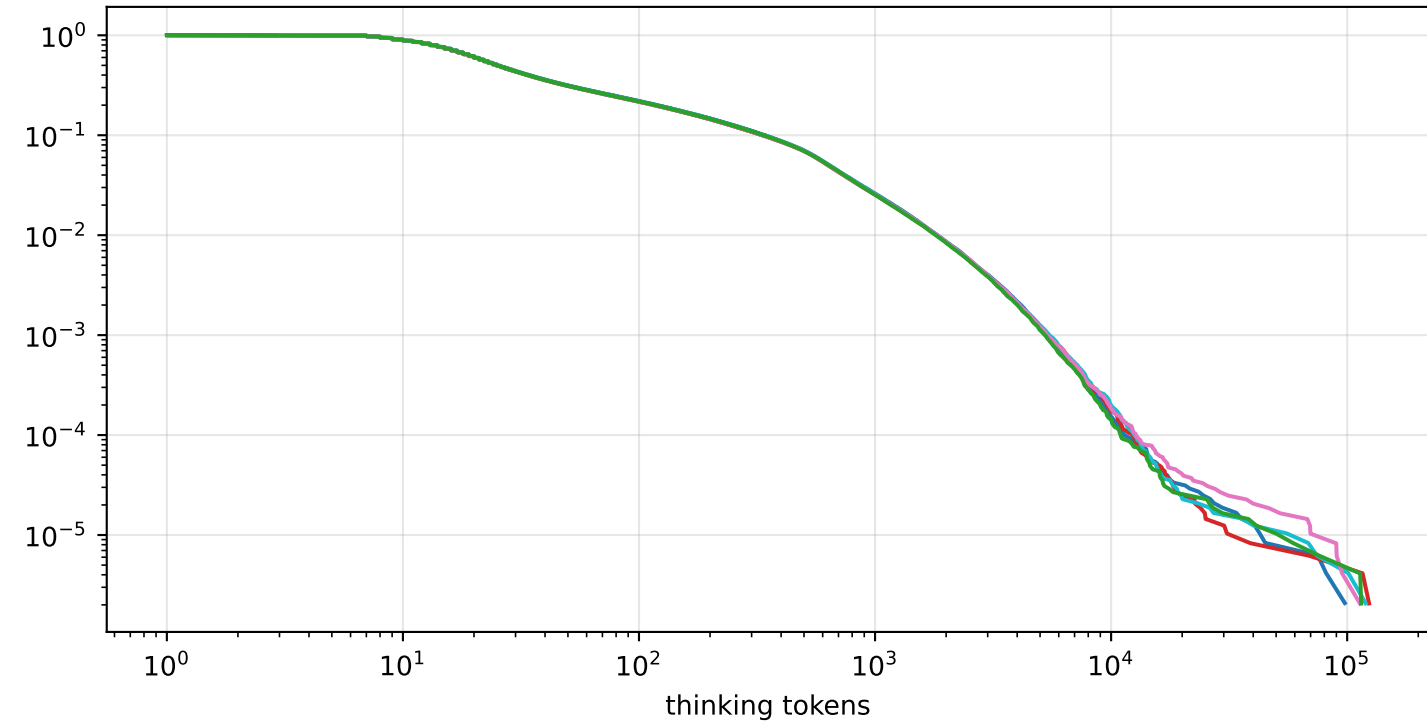


Reasoning length per turn

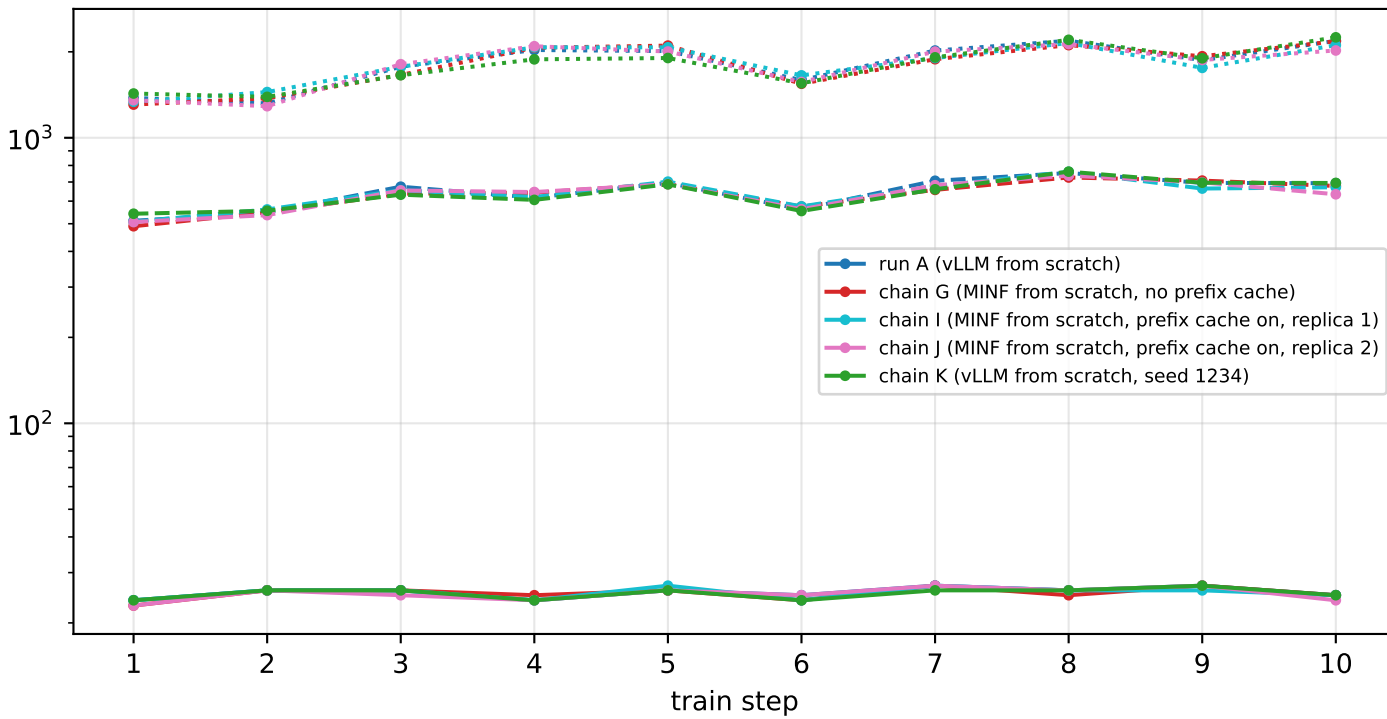
CDF of thinking tokens per assistant turn



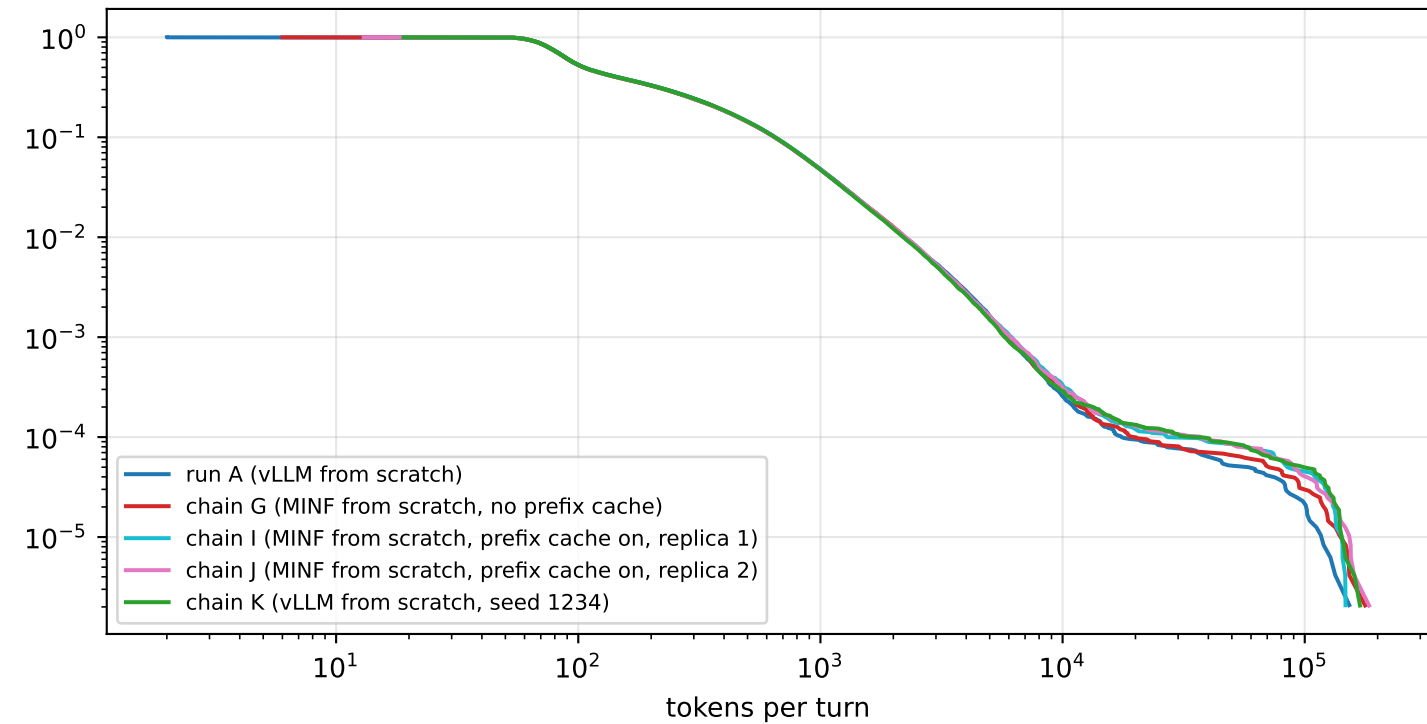
Tail: share of turns with thinking > x tokens



Per-step thinking-length percentiles: solid p50, dashed p95, dotted p99

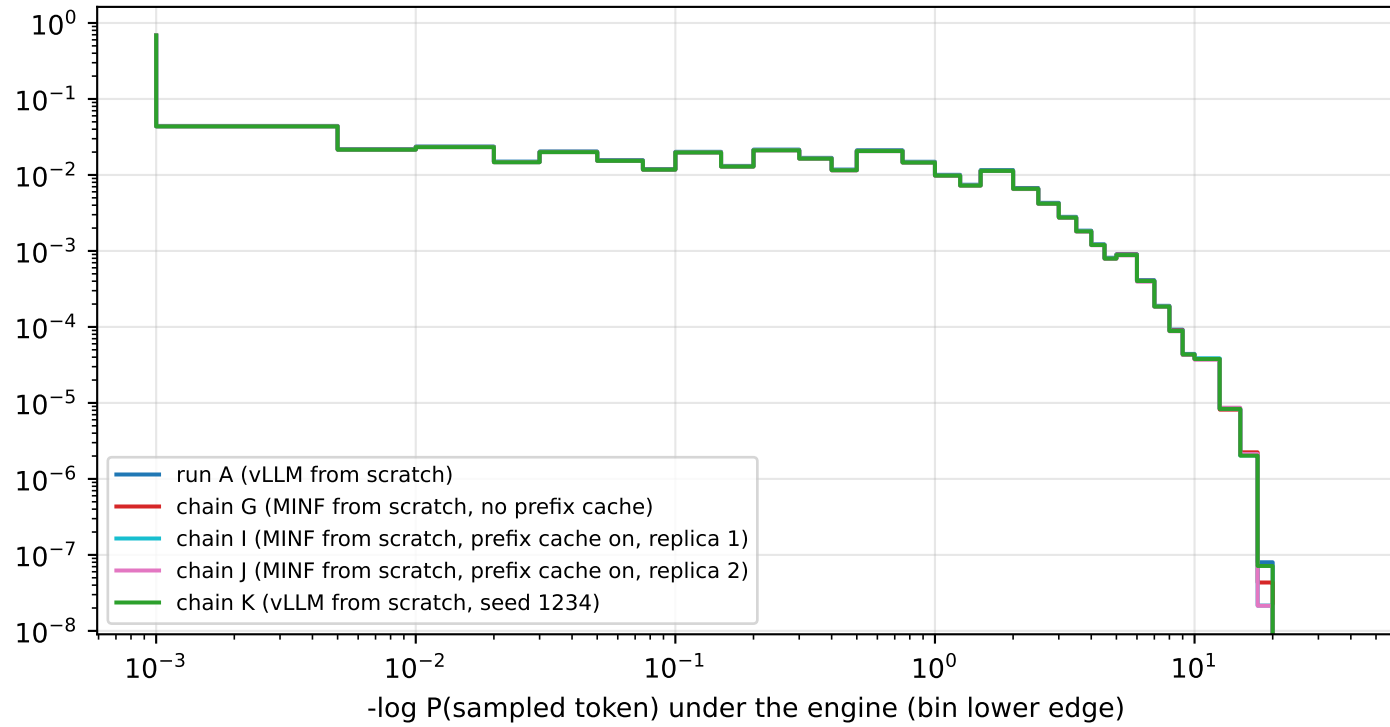


Tail of generated tokens per assistant turn (whole turn incl. tool call)

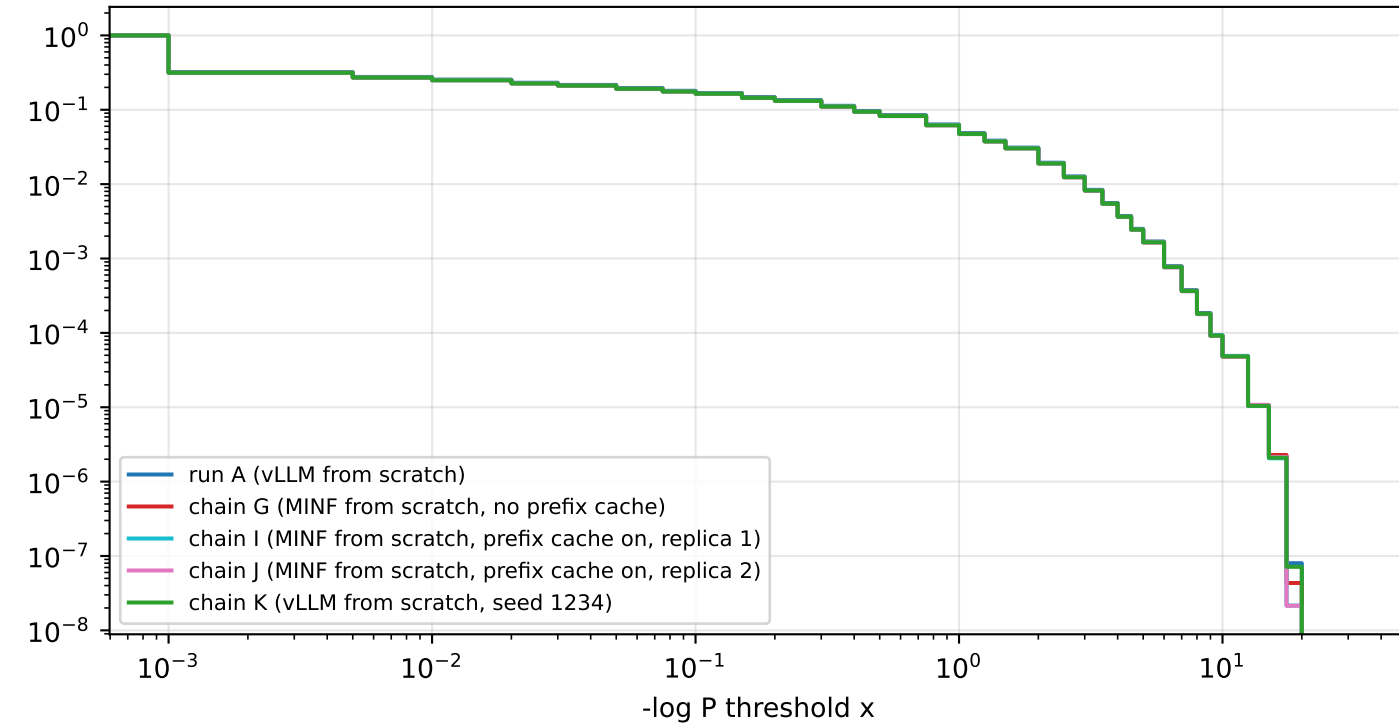


Token-level engine logprob distribution including the tail

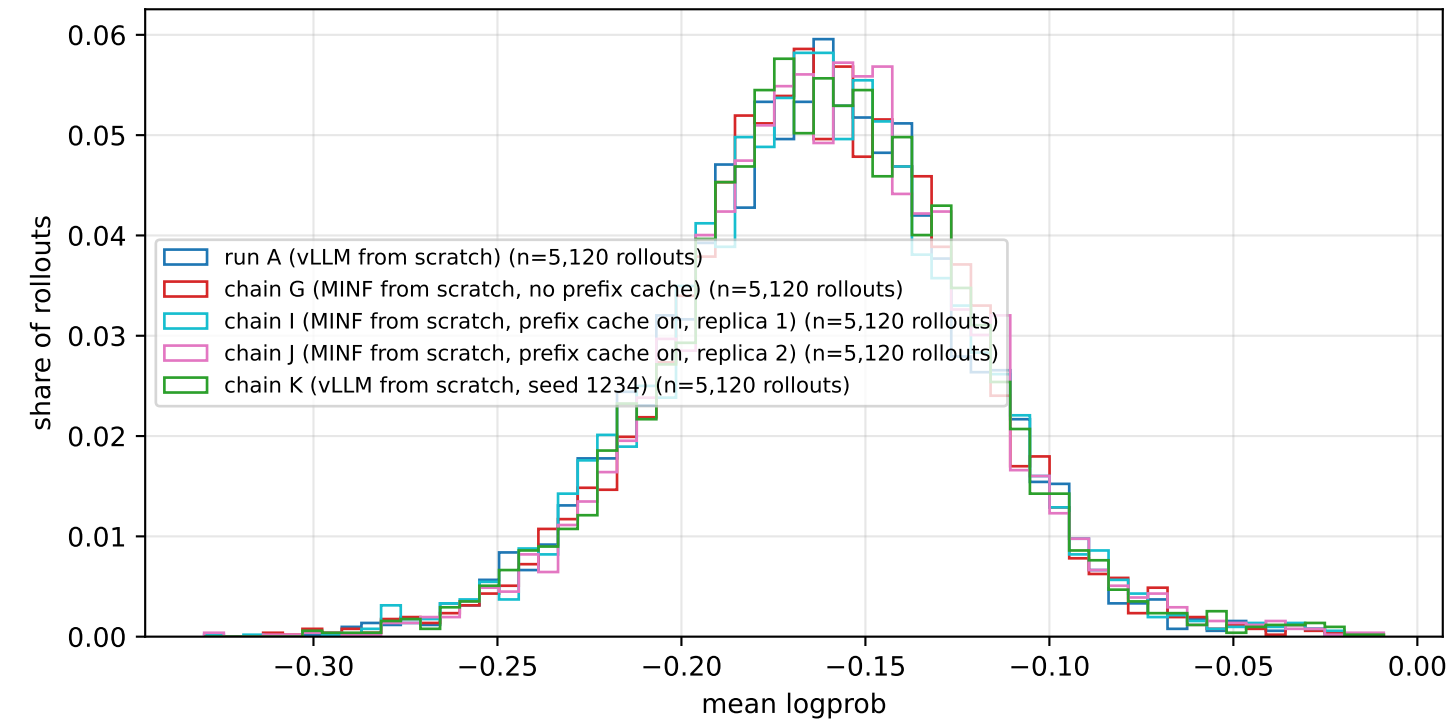
Engine logprob of sampled tokens, share per bin



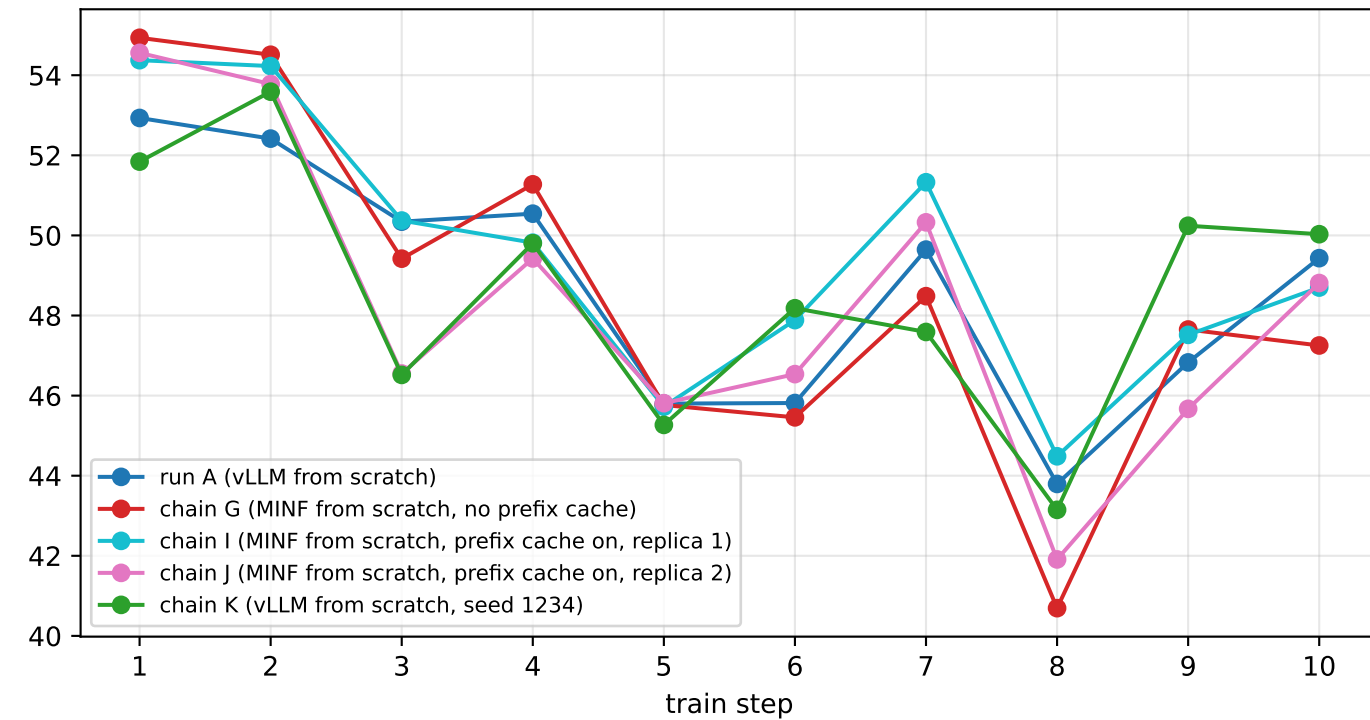
Tail: share of tokens with $-\log P \geq x$



Per-rollout mean engine logprob of generated tokens (share of rollouts)

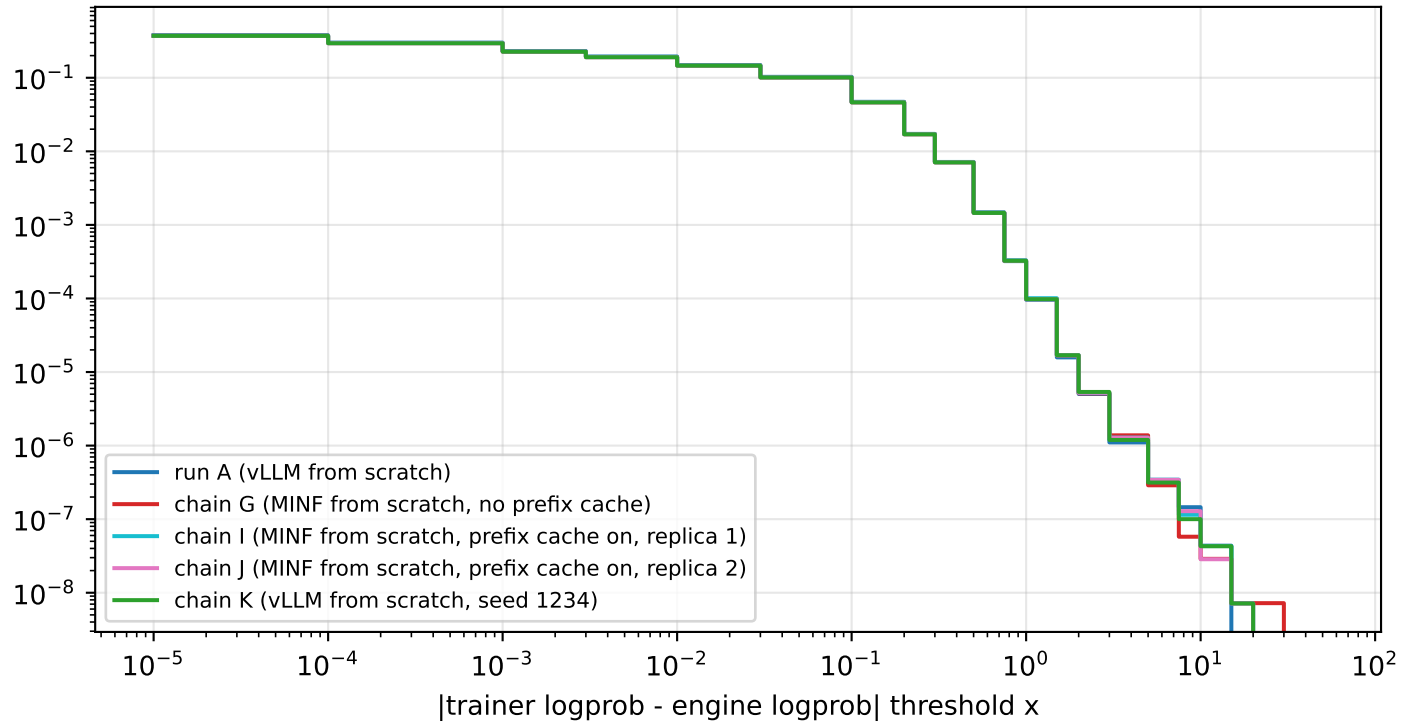


Tokens with engine logprob < -10 , per million generated tokens

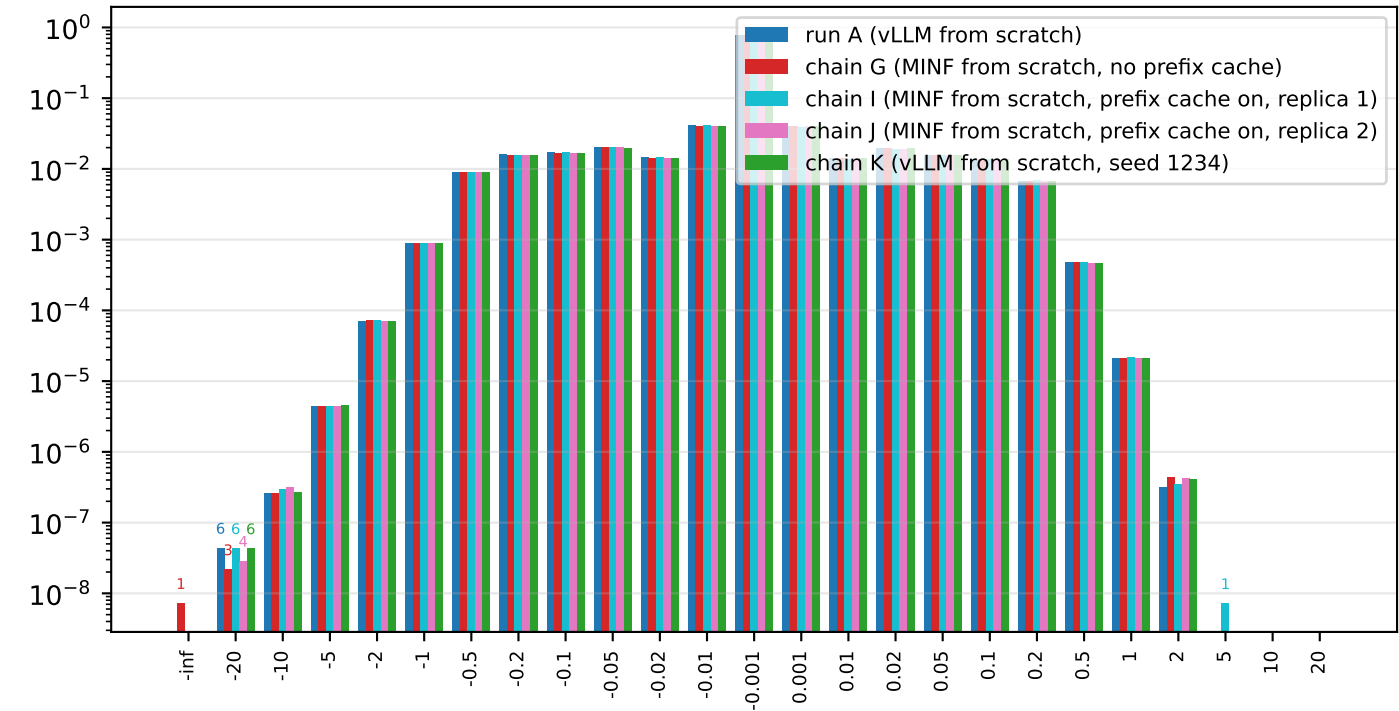


Trainer vs engine logprob mismatch

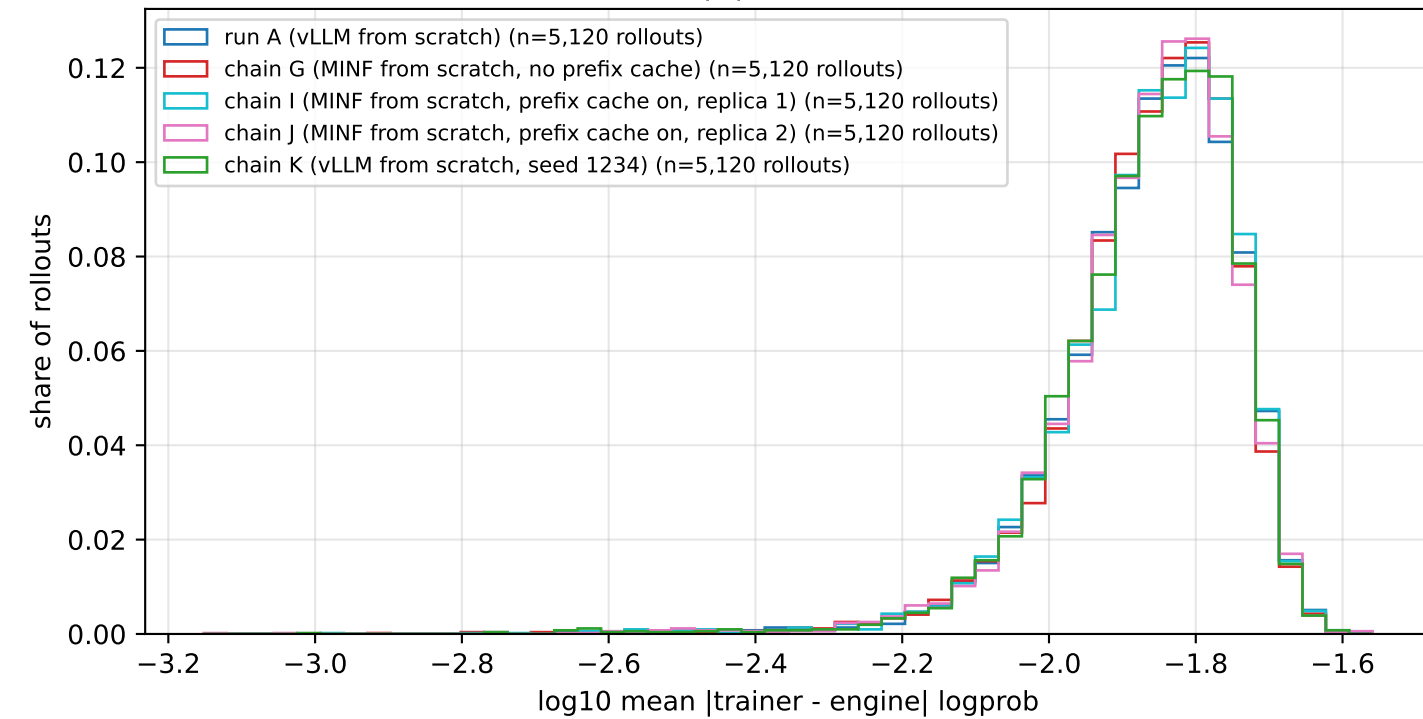
Tail: share of tokens with $|d| > x$



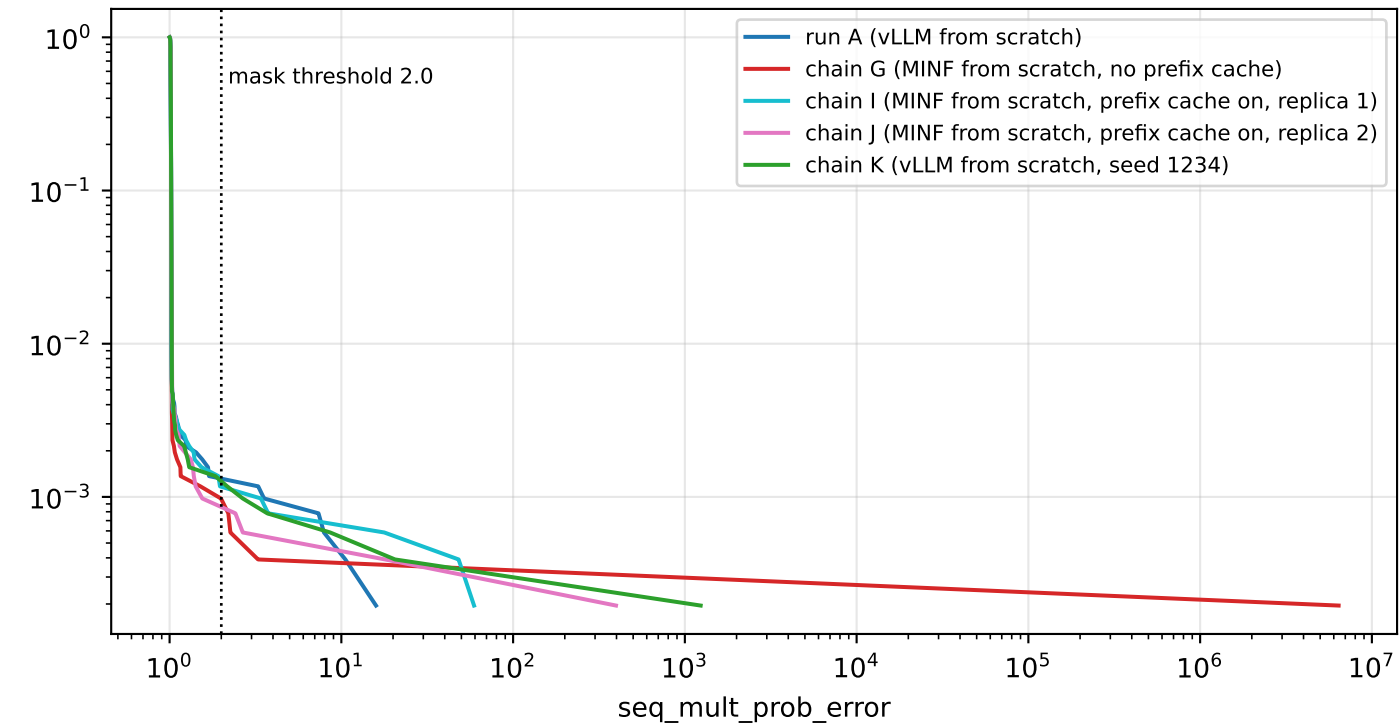
Signed $d = \text{trainer} - \text{engine logprob}$ (bin lower edges; leftmost bin = $d < -20$)



Per-rollout mean $|d|$ (log10), share of rollouts

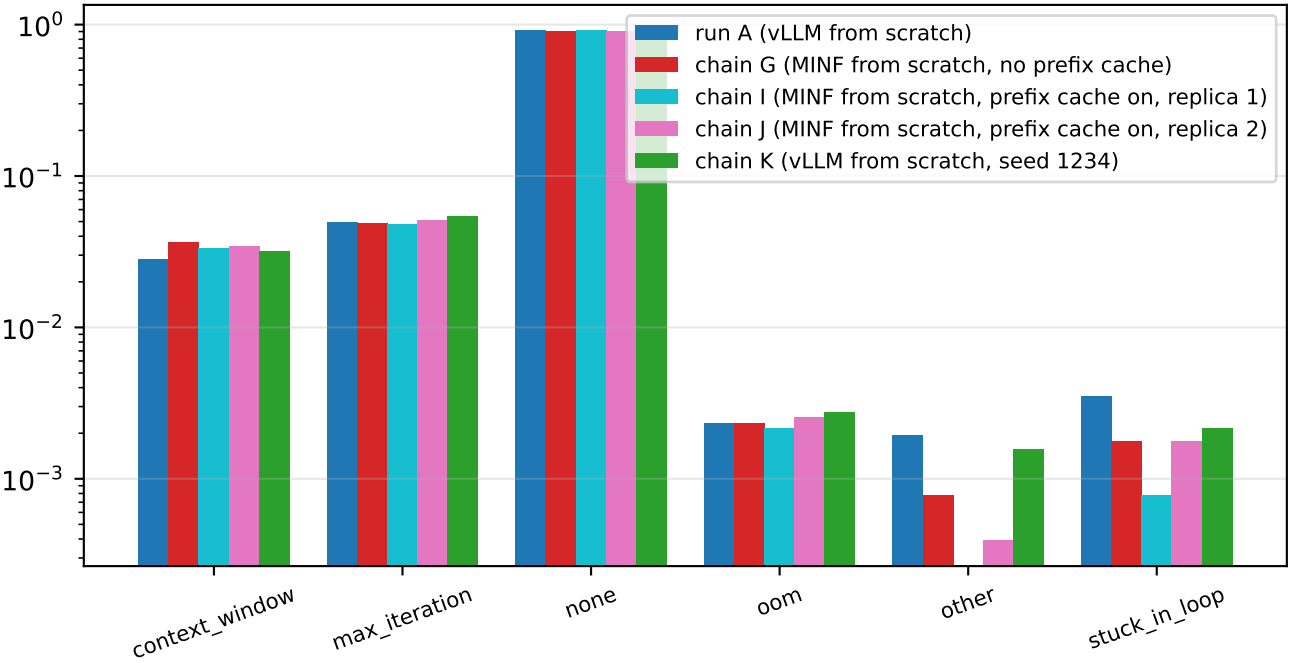


Tail of per-sequence multiplicative prob error

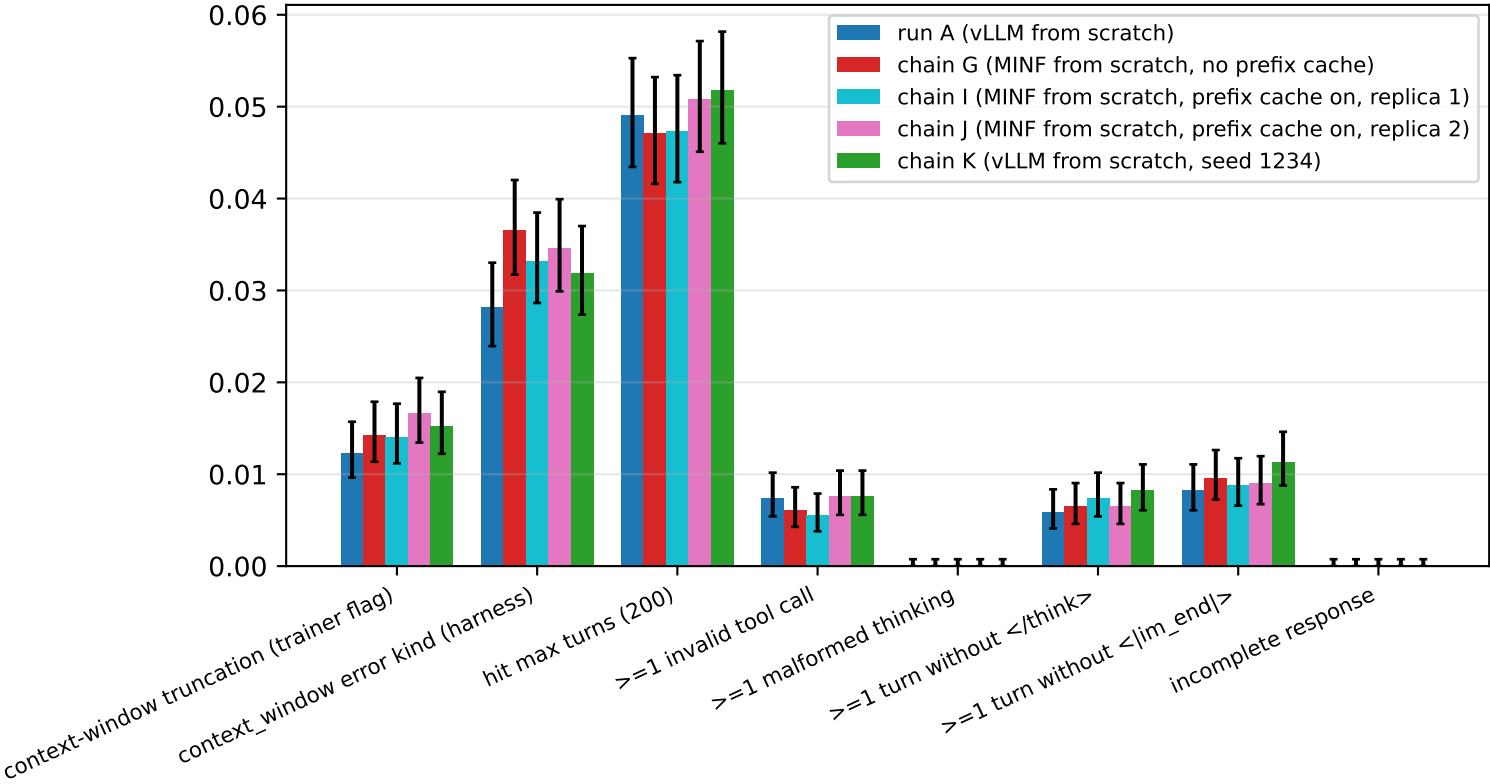


Invalid tool calls, malformed thinking, generation limits and termination kinds

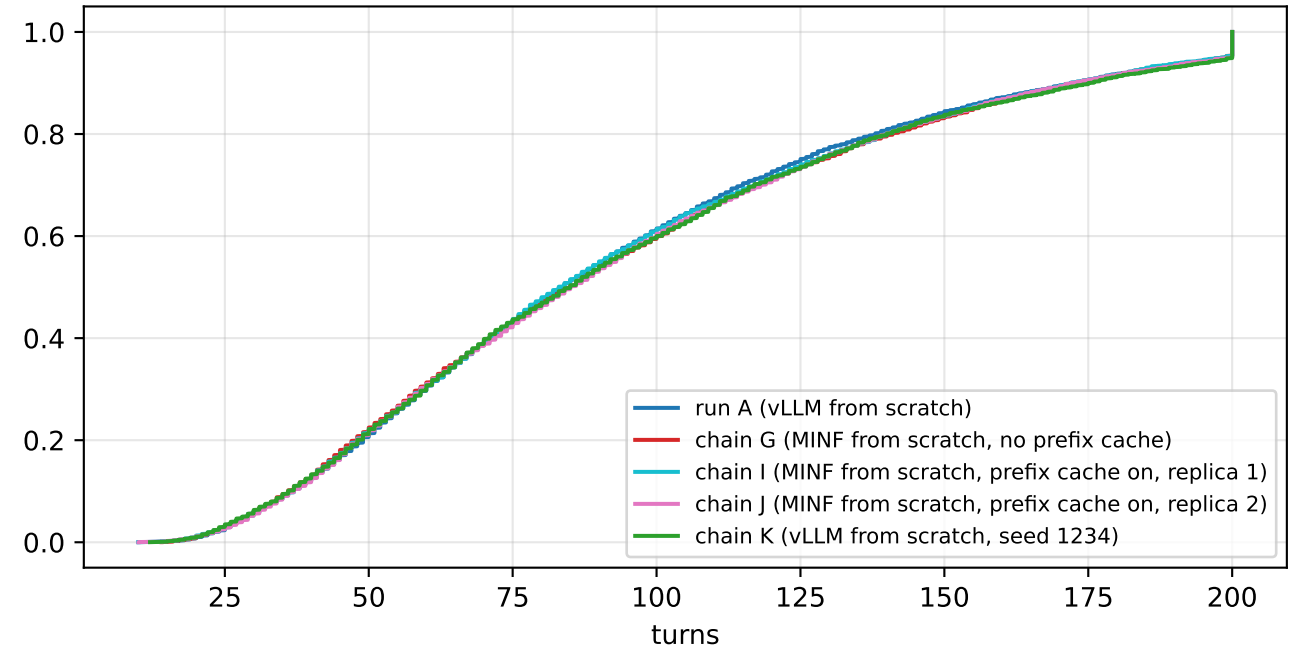
Agent termination / error kind (share of rollouts)



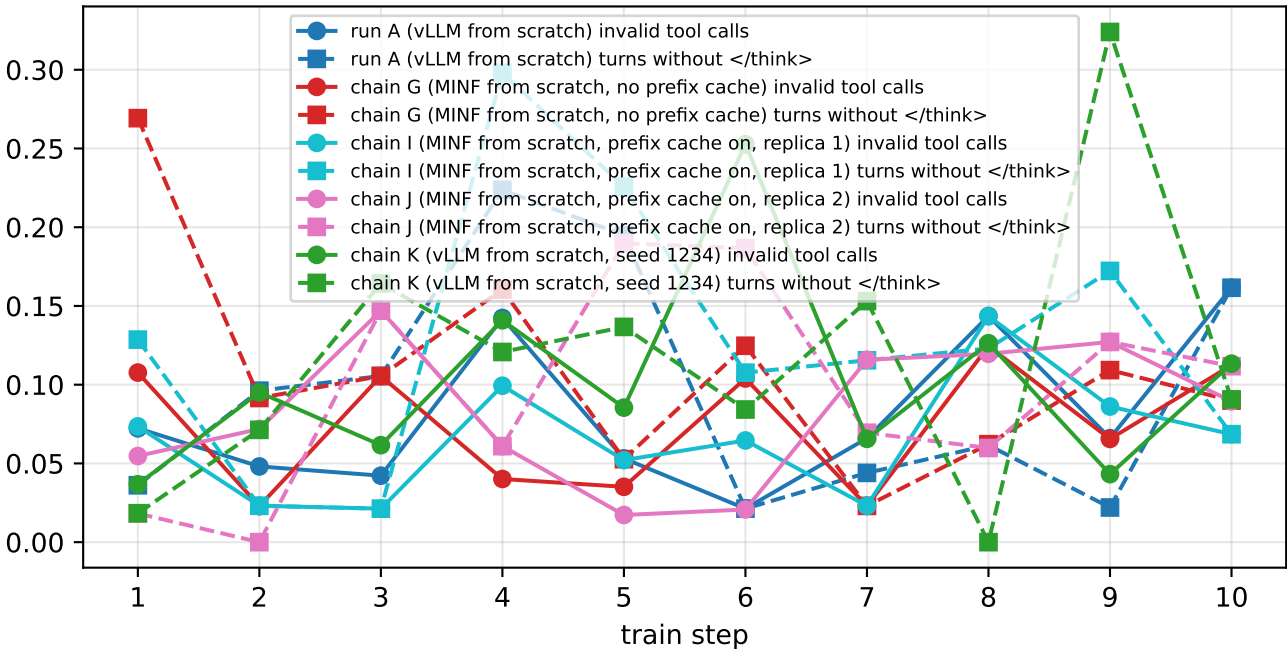
Limit and failure rates per rollout (95% Wilson CI)



CDF of assistant turns per rollout

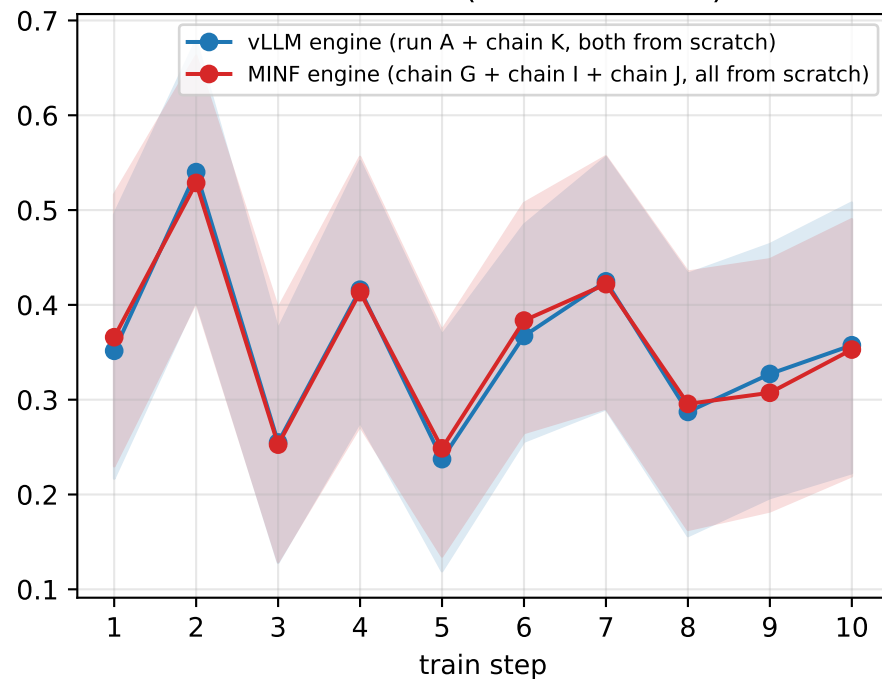


Per-turn rates, per 1,000 assistant turns

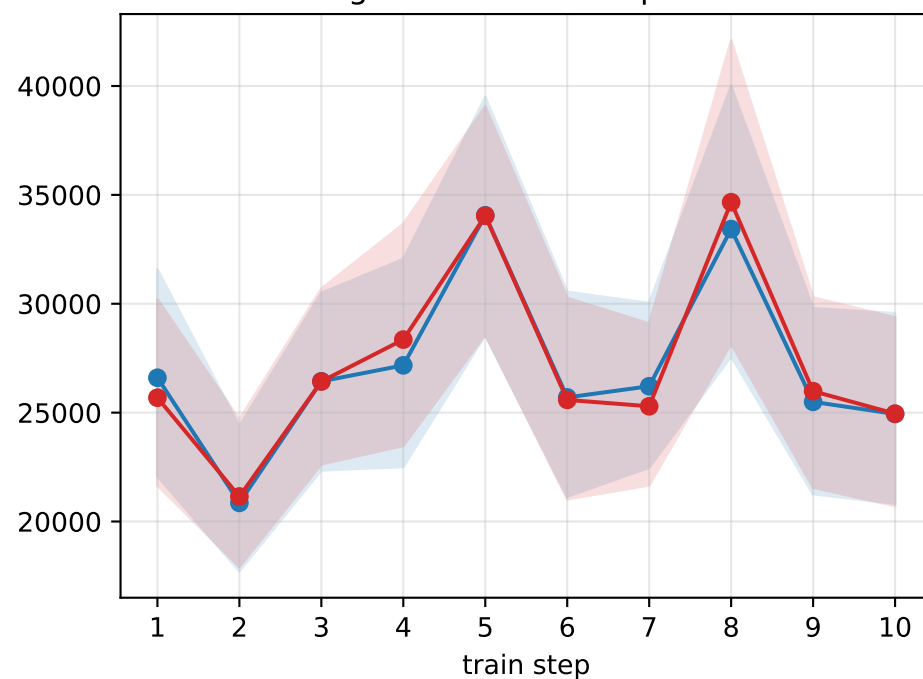


[engine-pooled] Per-step overview, steps 1-10 (bands = 95% prompt-cluster bootstrap CI of the step mean)

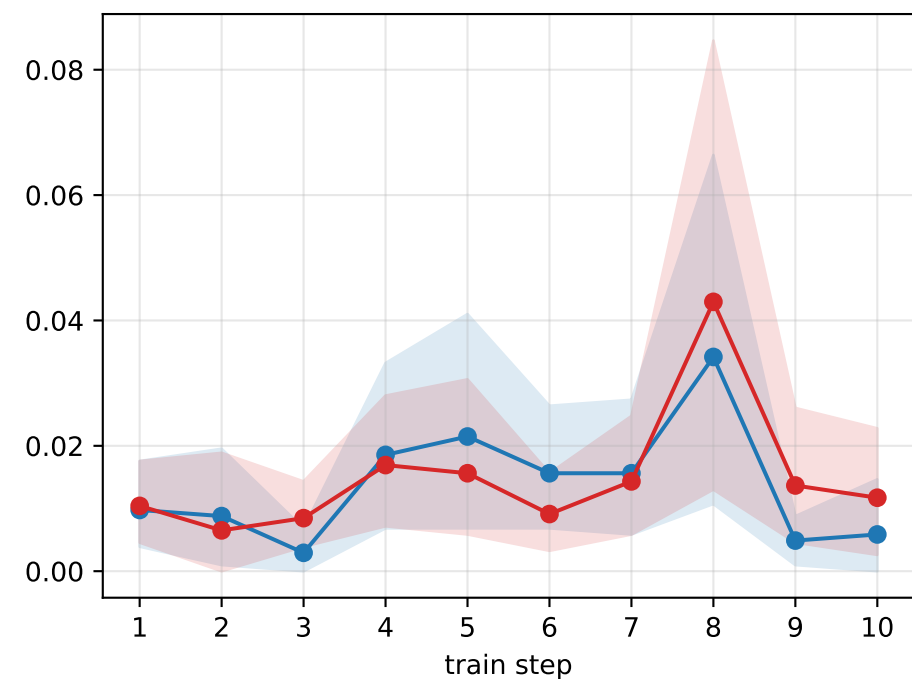
mean reward (share resolved)



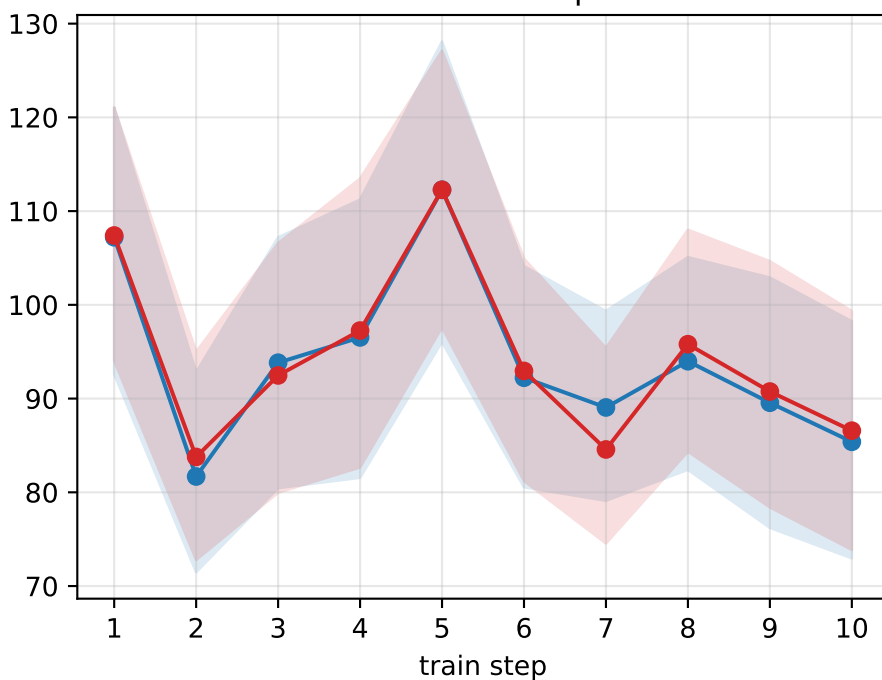
mean generated tokens per rollout



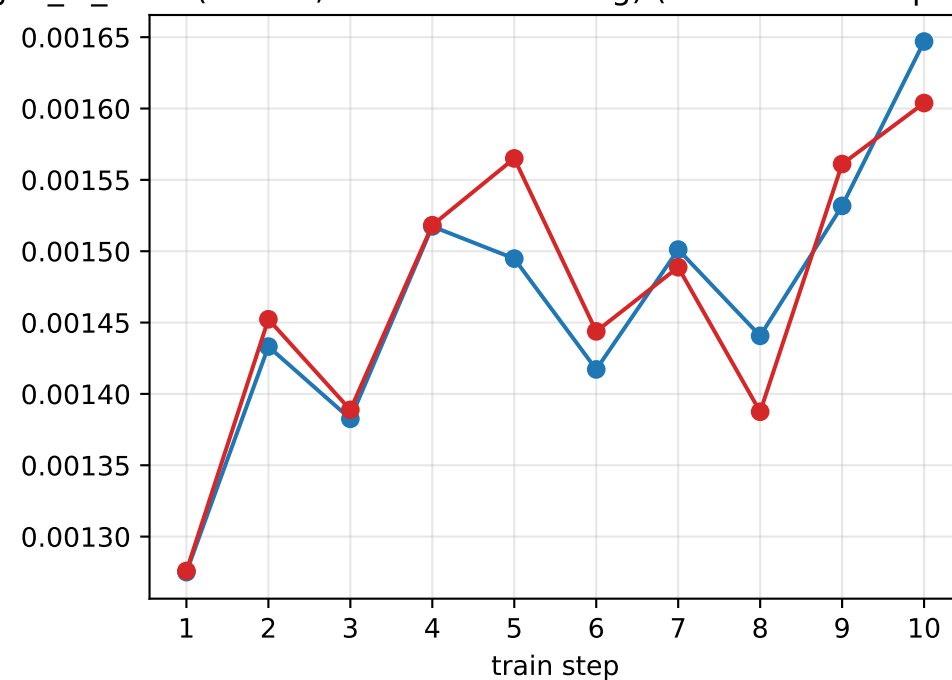
context-window truncation rate



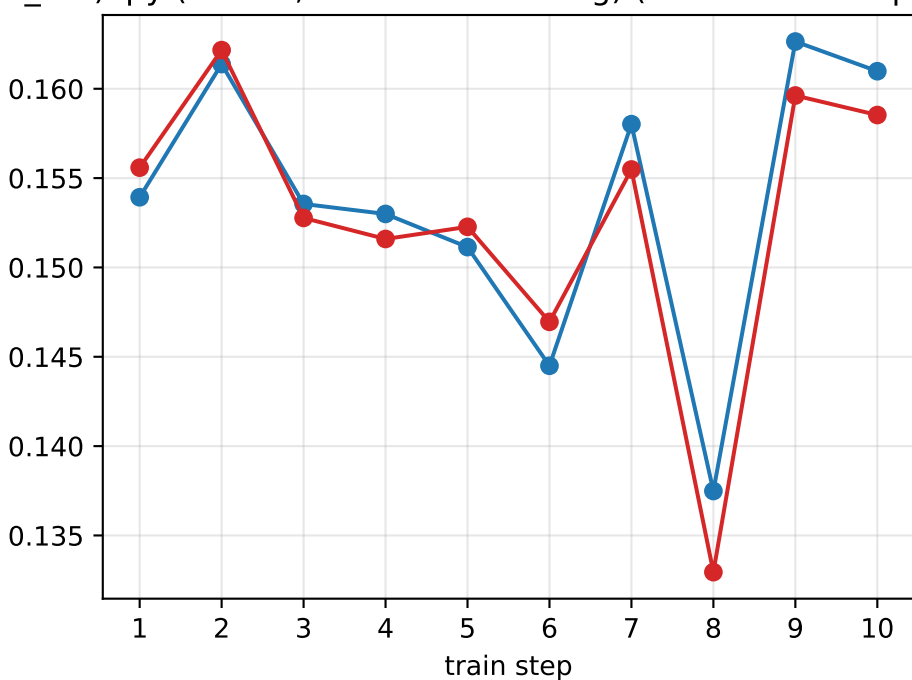
mean assistant turns per rollout



gen_kl_error (trainer, from controller log) (mean over the population)

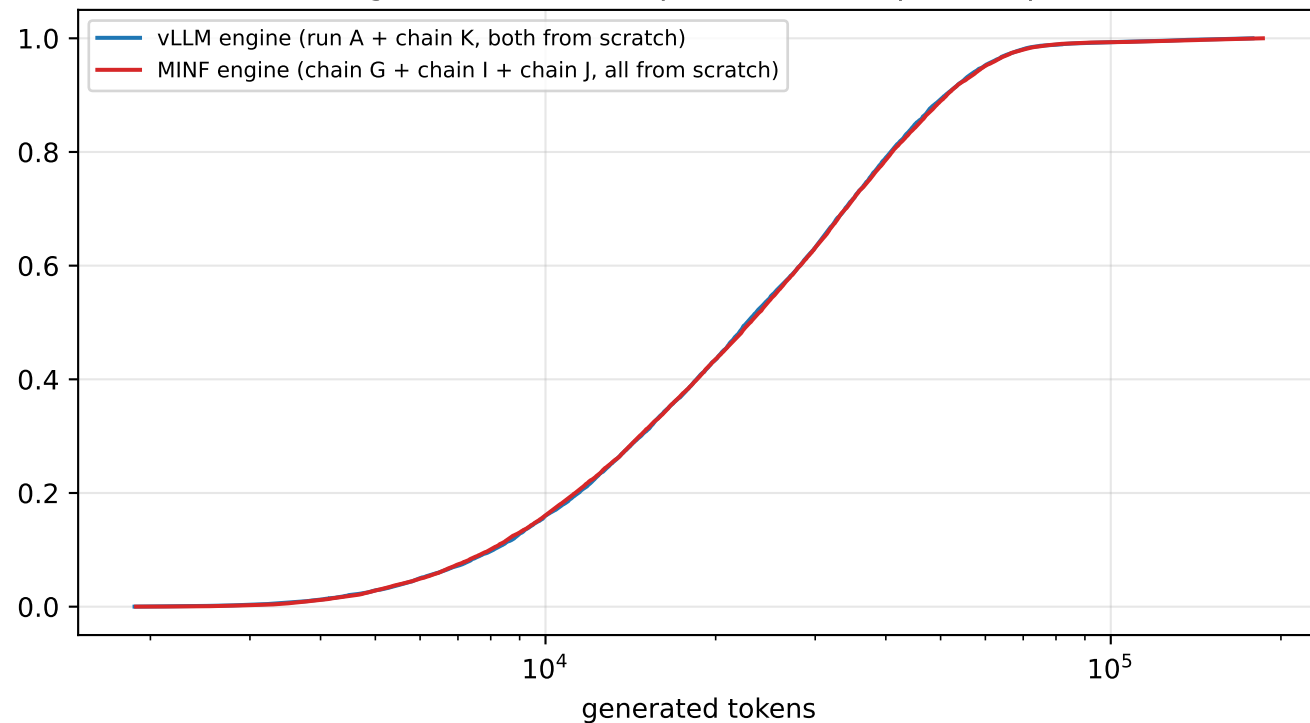


pop_entropy (trainer, from controller log) (mean over the population)

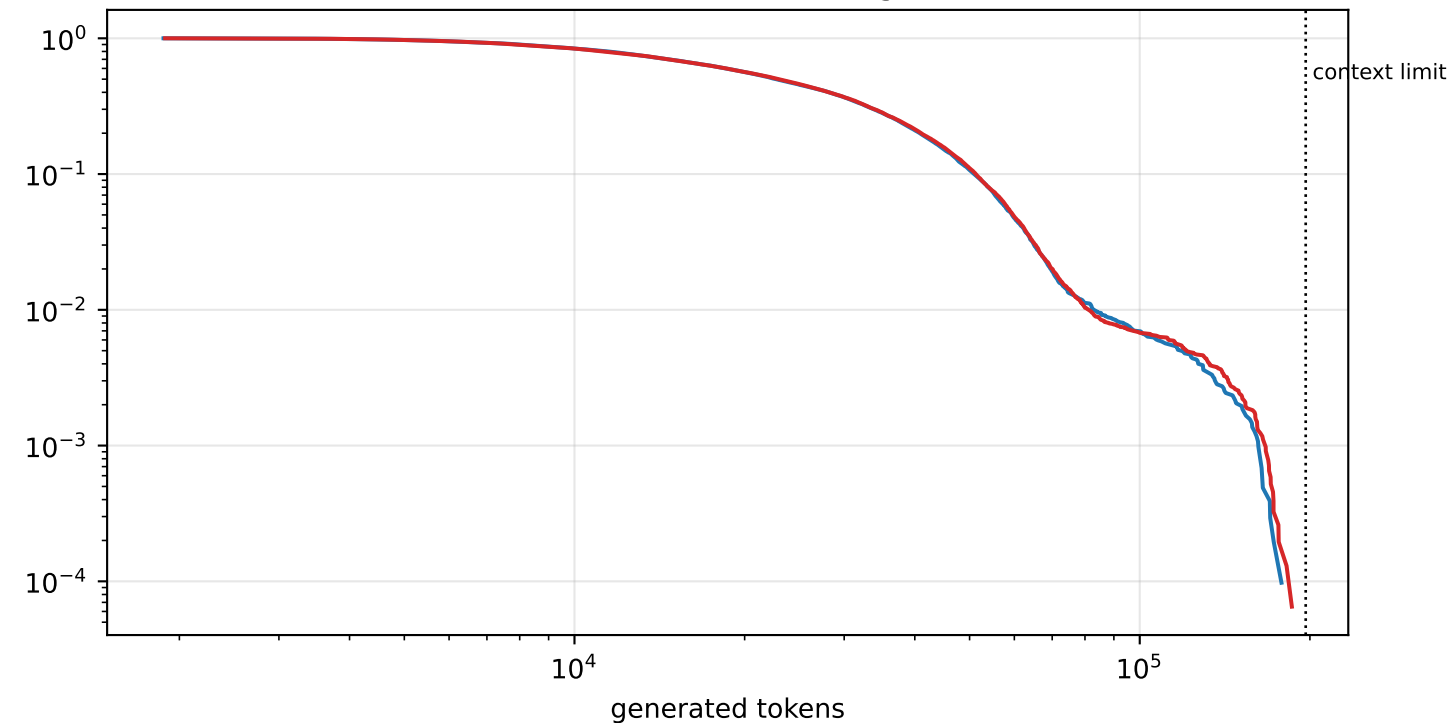


[engine-pooled] Generation length distribution including the tail

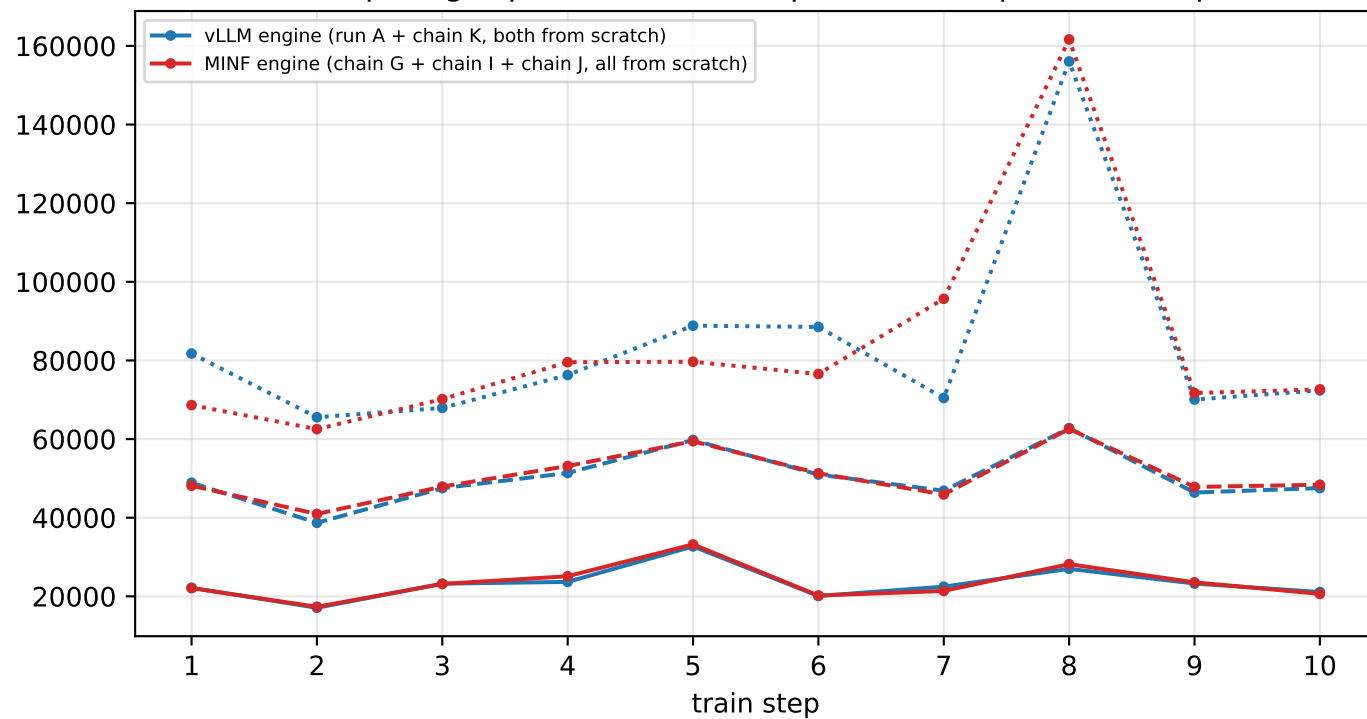
CDF of generated tokens per rollout (steps 1-10 pooled)



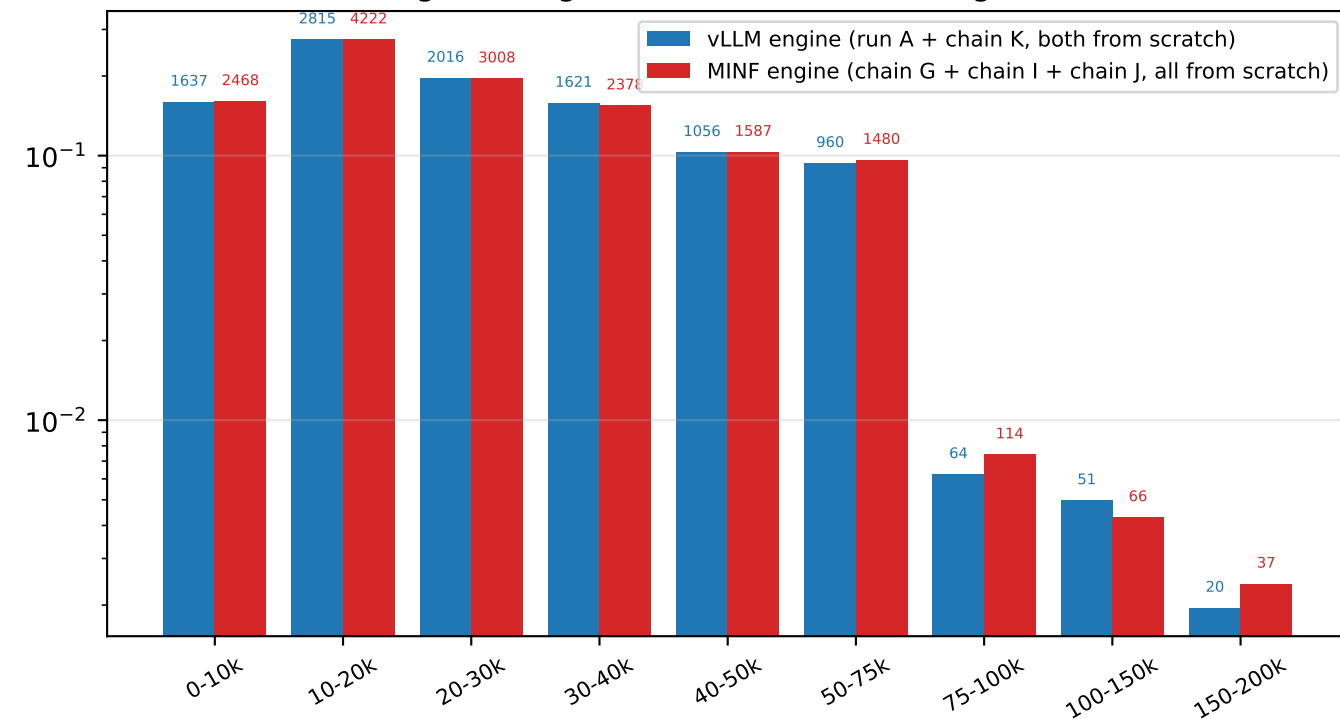
Tail: survival function P(length > x)



Per-step length percentiles: solid p50, dashed p90, dotted p99

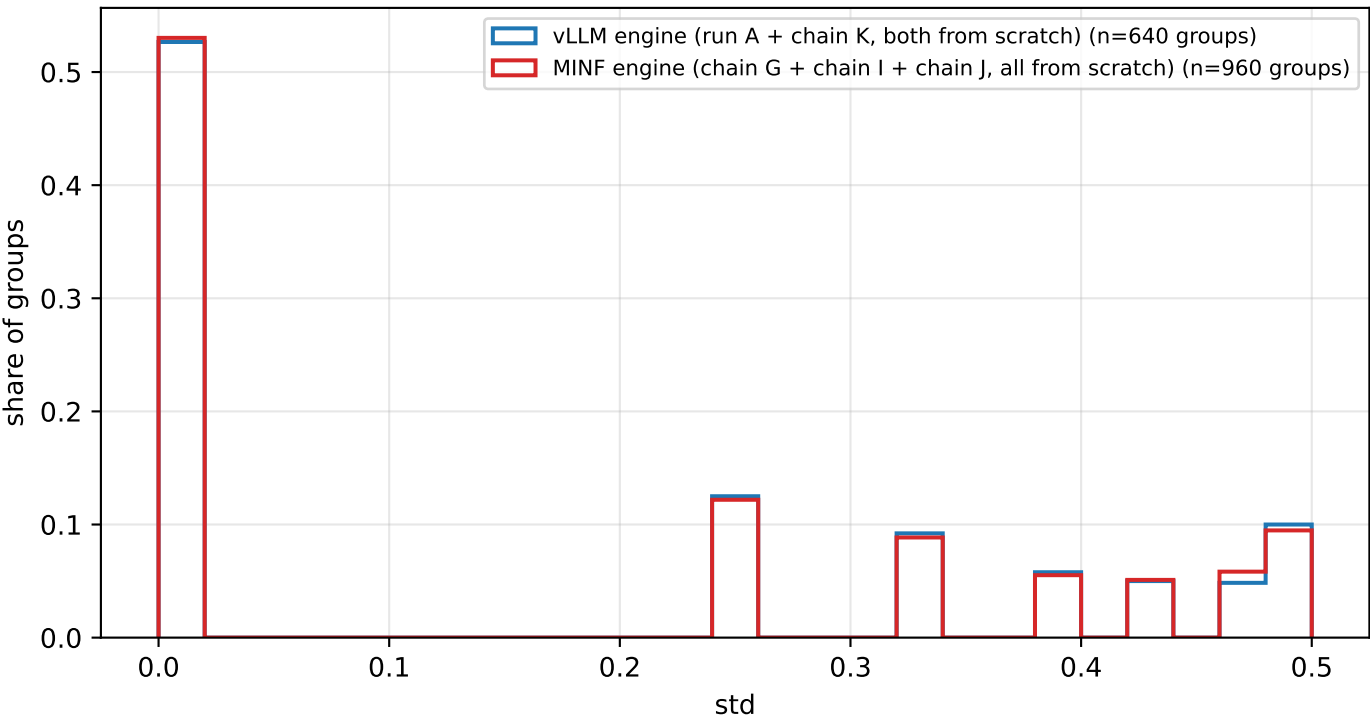


Length histogram (share of rollouts, log scale)

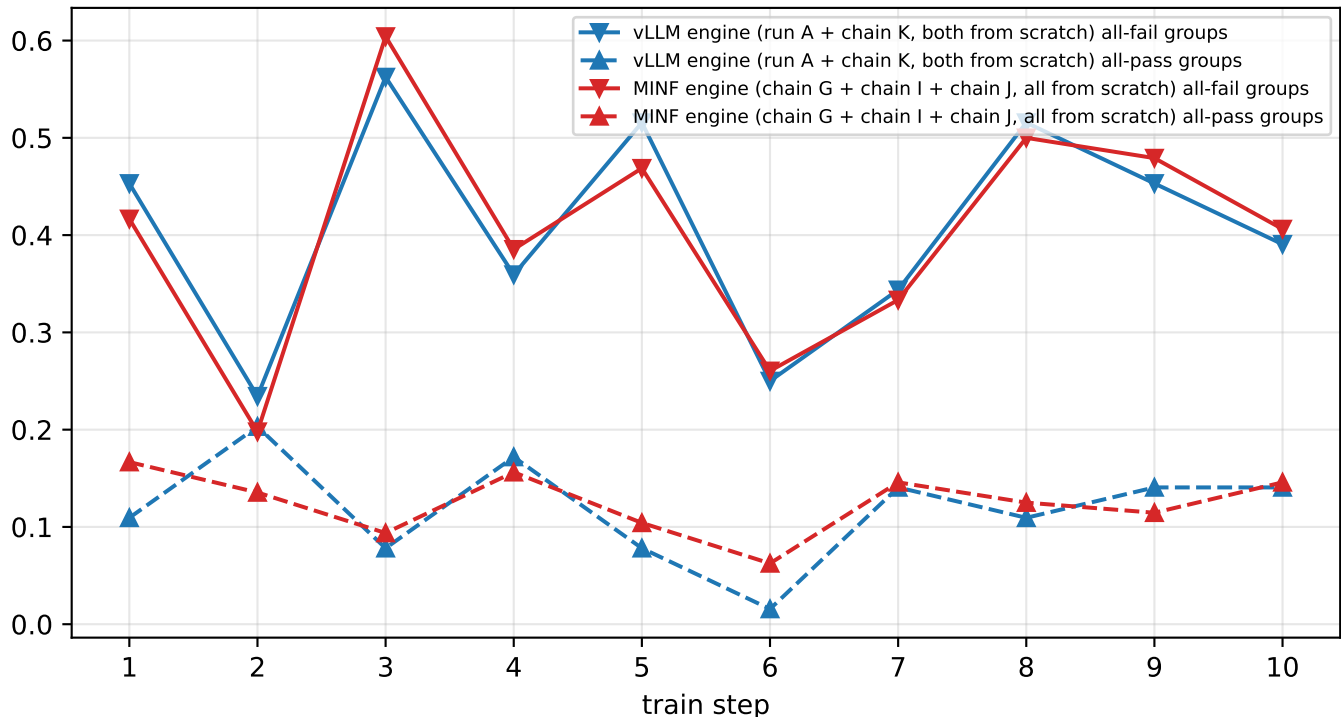


[engine-pooled] Reward and GRPO group structure

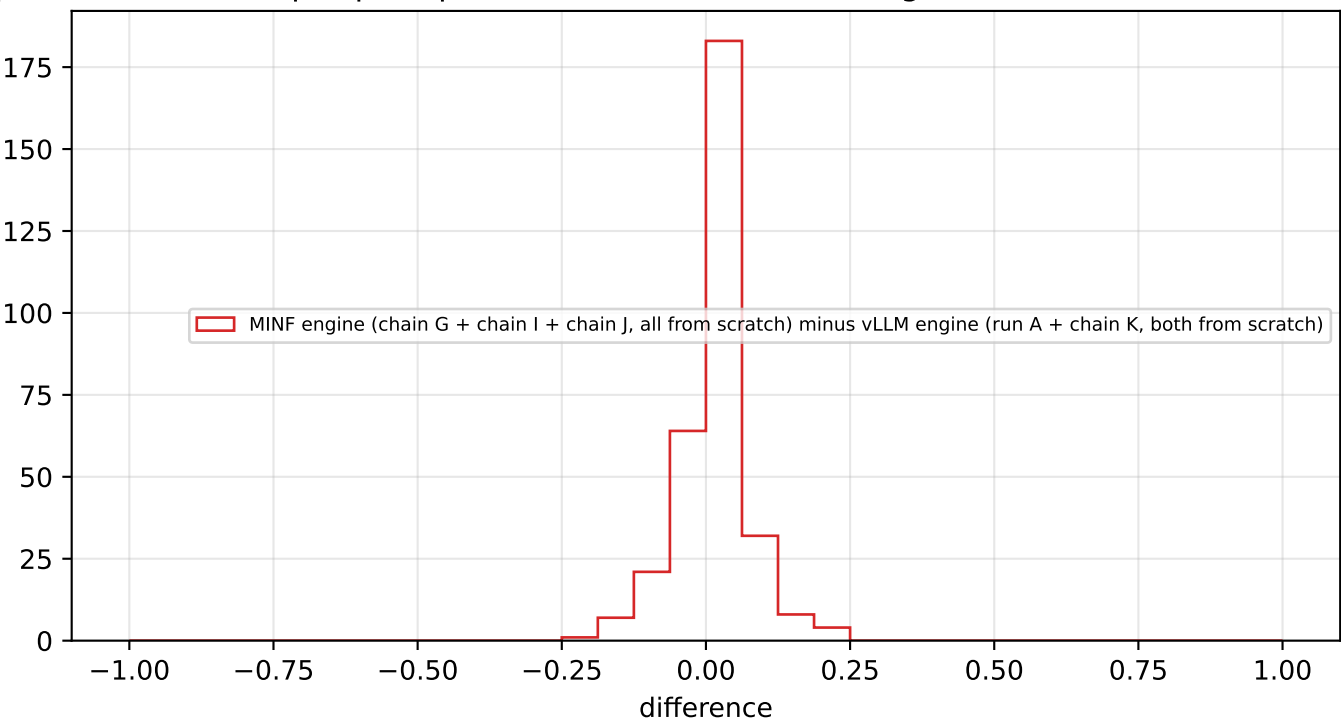
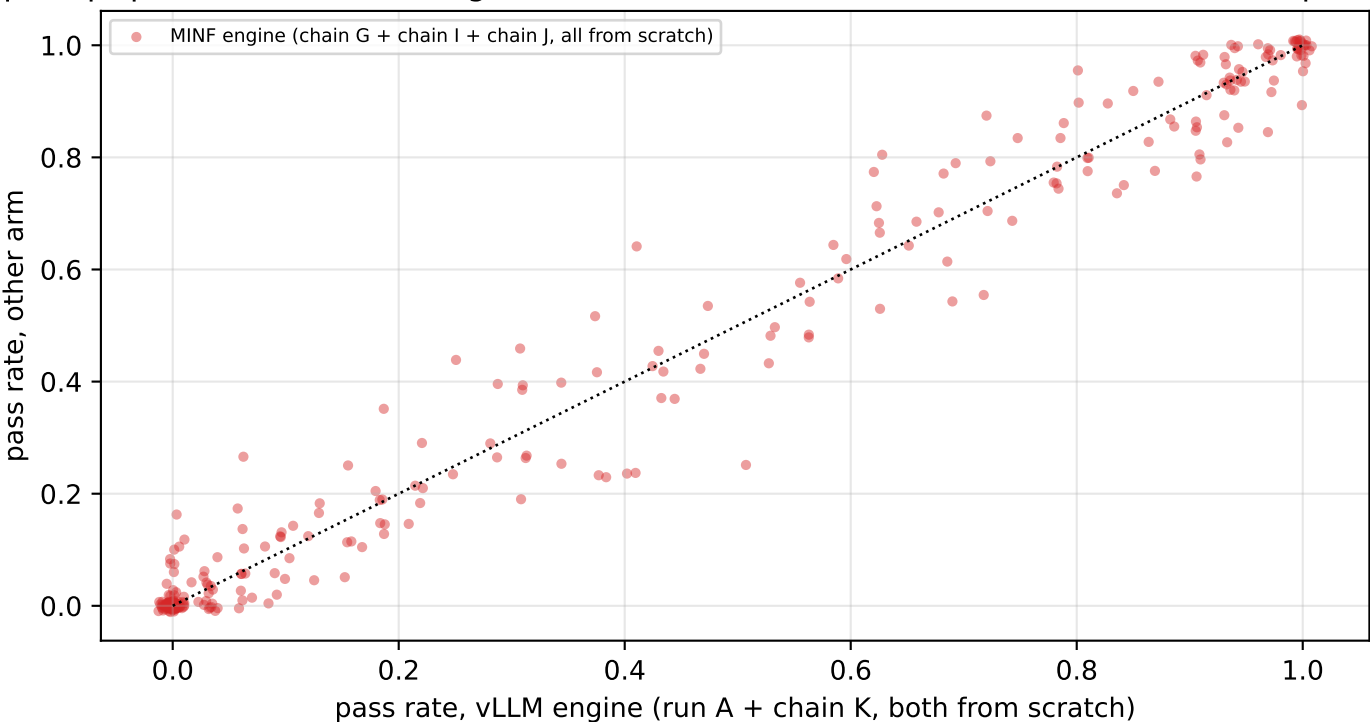
Within-group reward std (16 rollouts per group, ddof=0)



Share of degenerate groups (zero advantage signal)

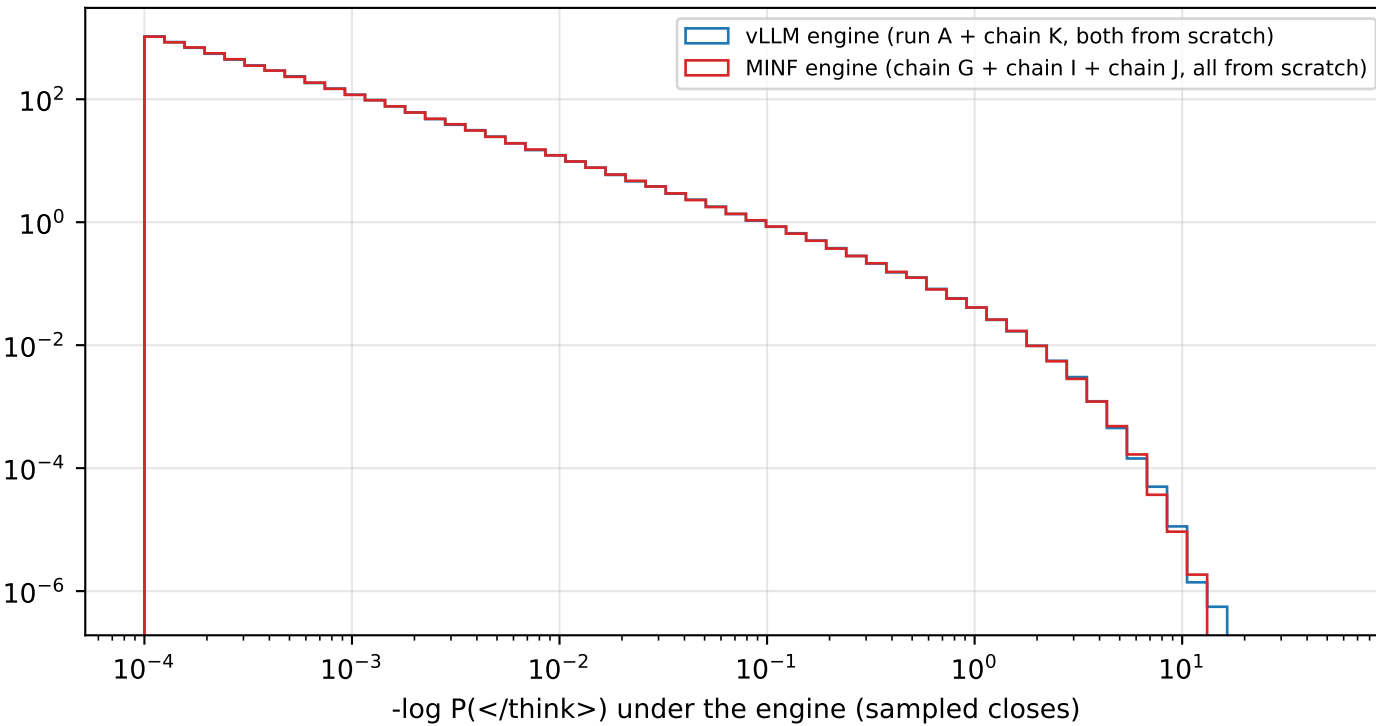


per-prompt pass rate vs vLLM engine (run A + chain K, both from scratch) (matched prompts) per-prompt pass rate difference per prompt, other arm minus vLLM engine (run A + chain K, both from scratch)

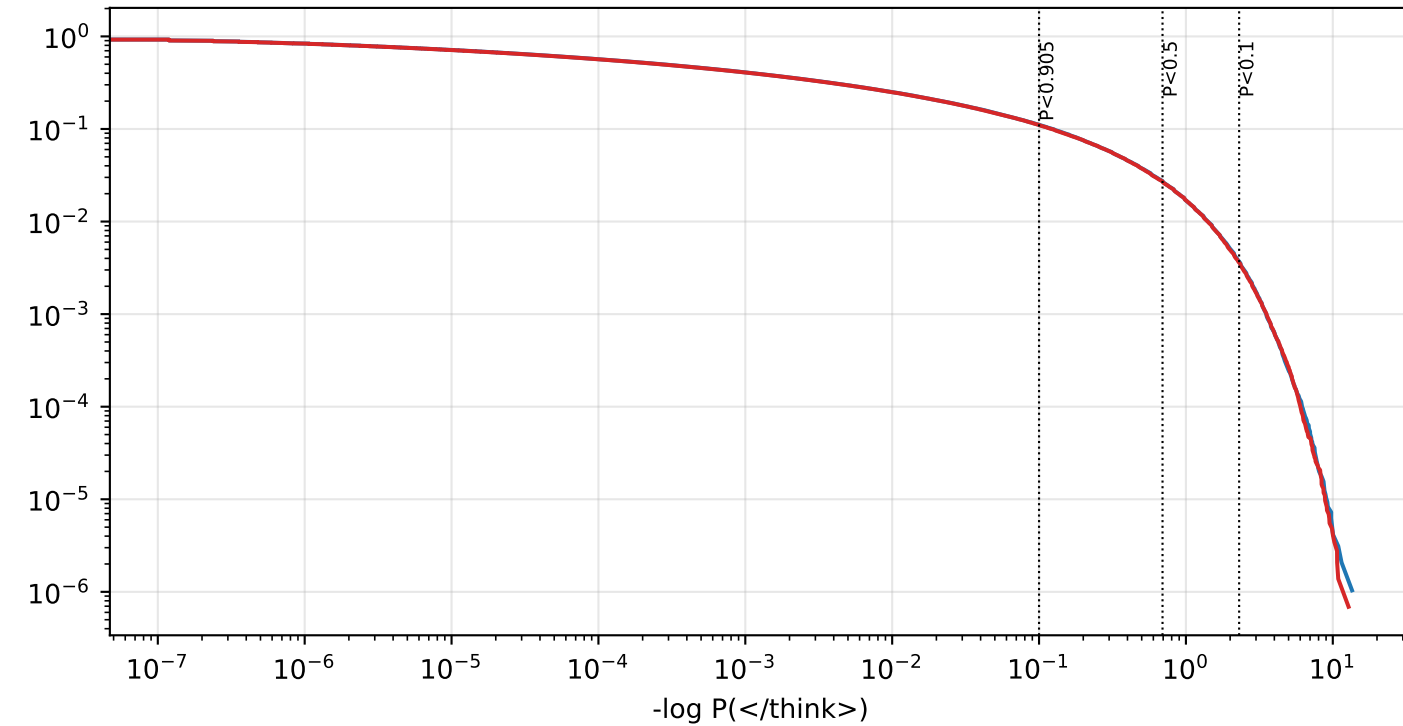


[engine-pooled] End-of-thinking token: engine probability of the sampled </think>

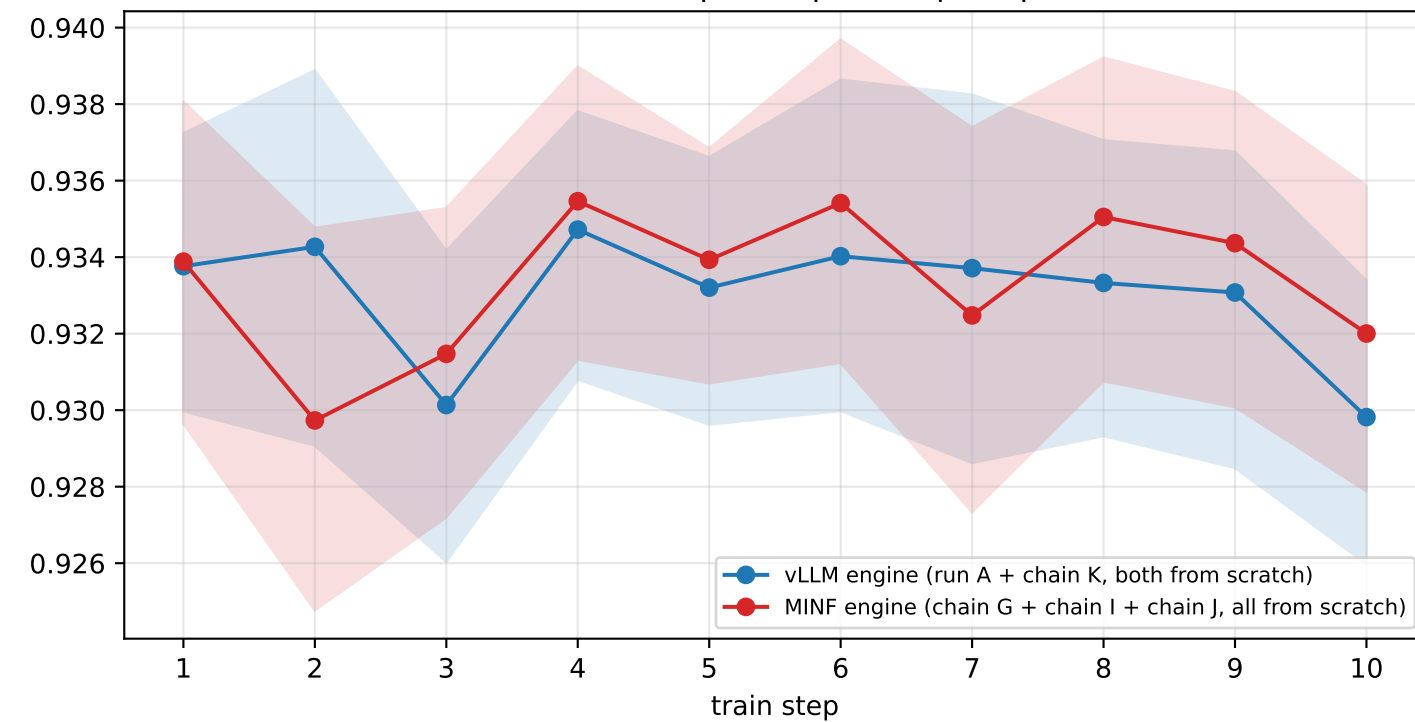
Distribution of the engine's </think> logprob



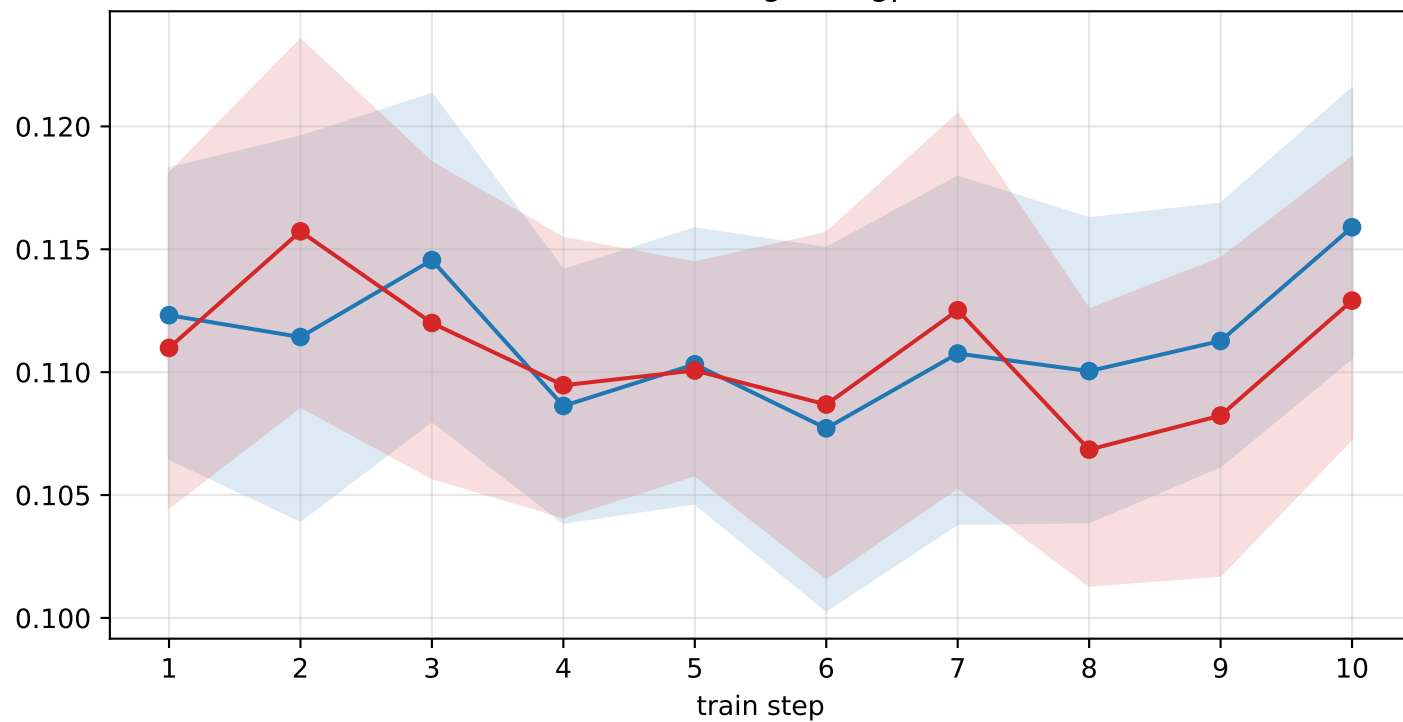
Tail: share of closes with $-\log P > x$



Geo-mean $P(</think>)$ per step (95% prompt-cluster CI)

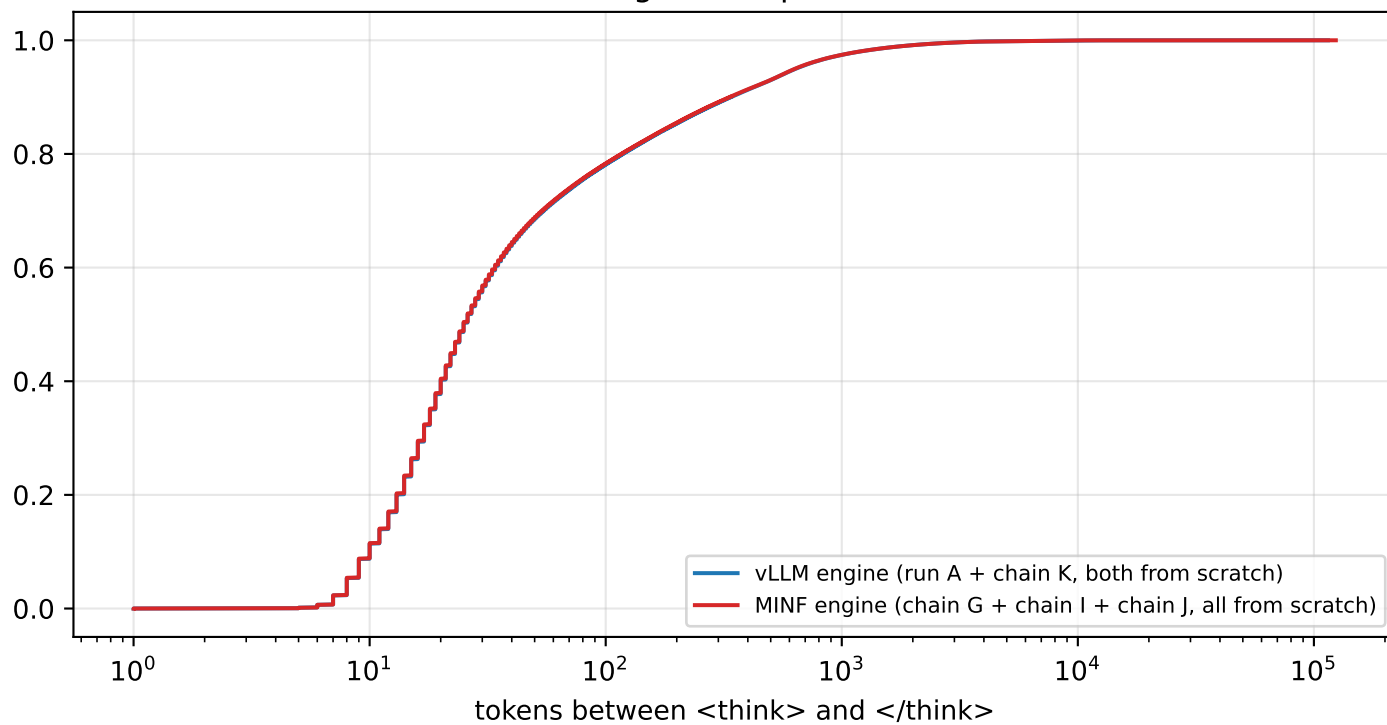


Low-confidence close share (engine logprob < -0.1 , $P < 0.905$)

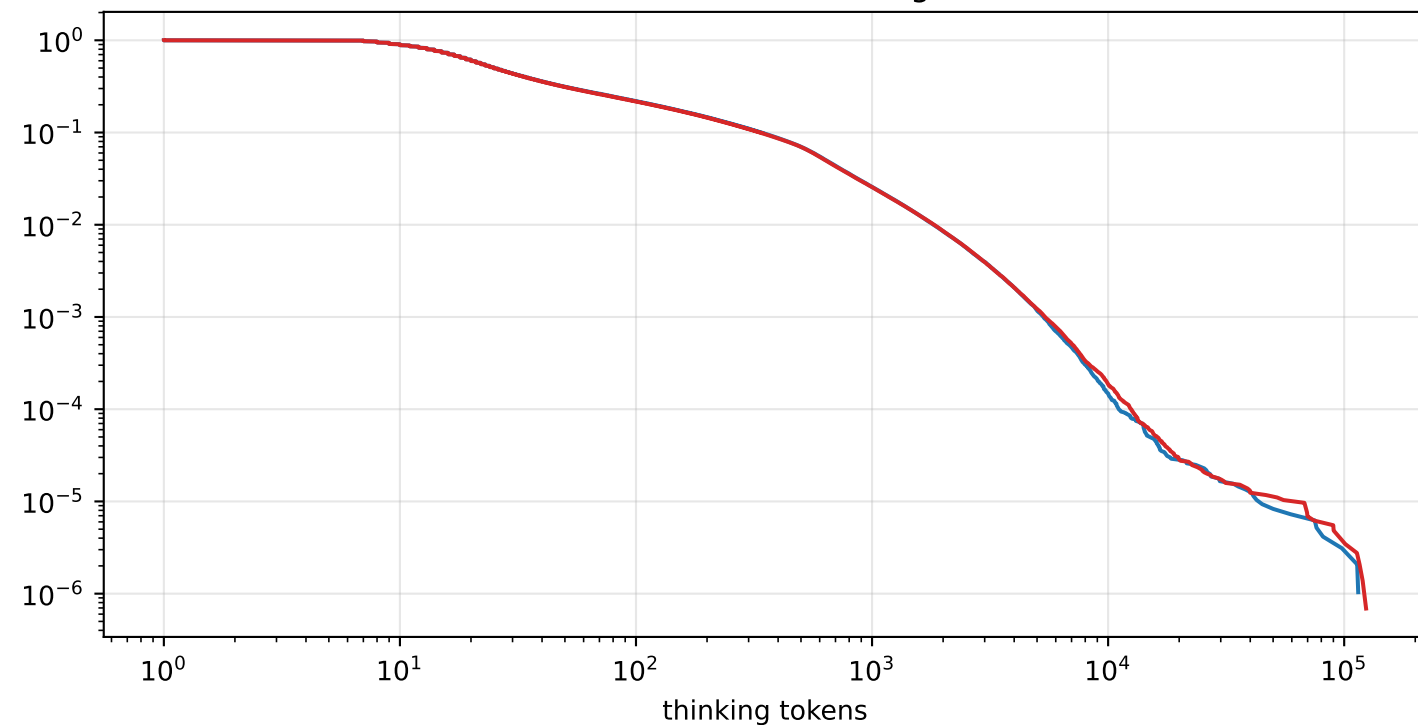


[engine-pooled] Reasoning length per turn

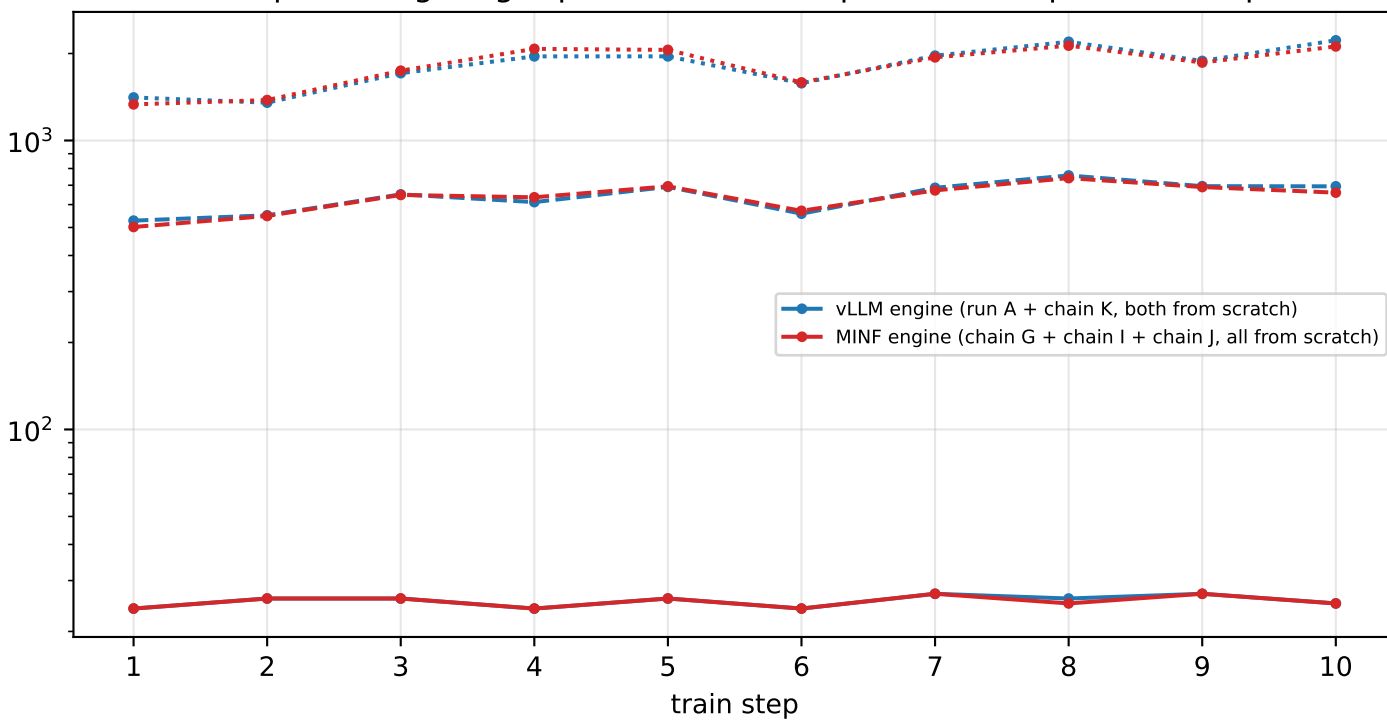
CDF of thinking tokens per assistant turn



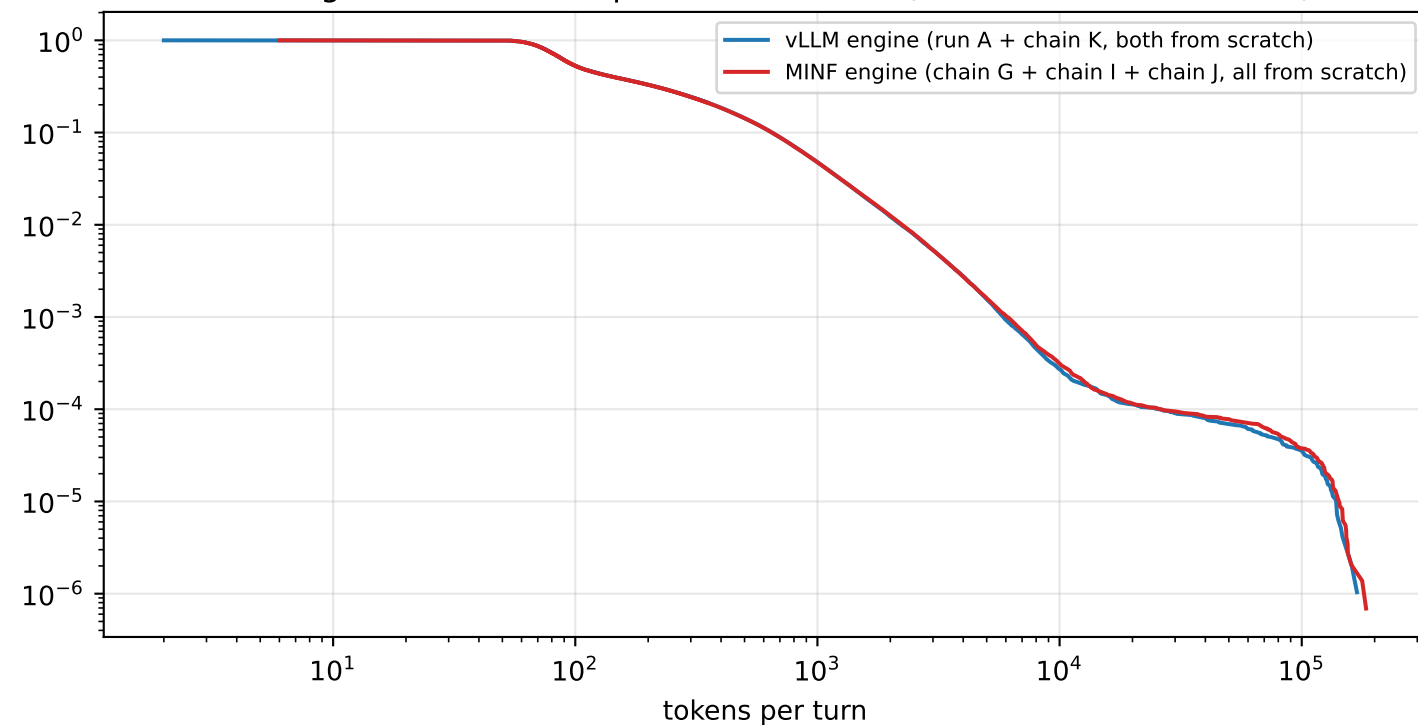
Tail: share of turns with thinking > x tokens



Per-step thinking-length percentiles: solid p50, dashed p95, dotted p99

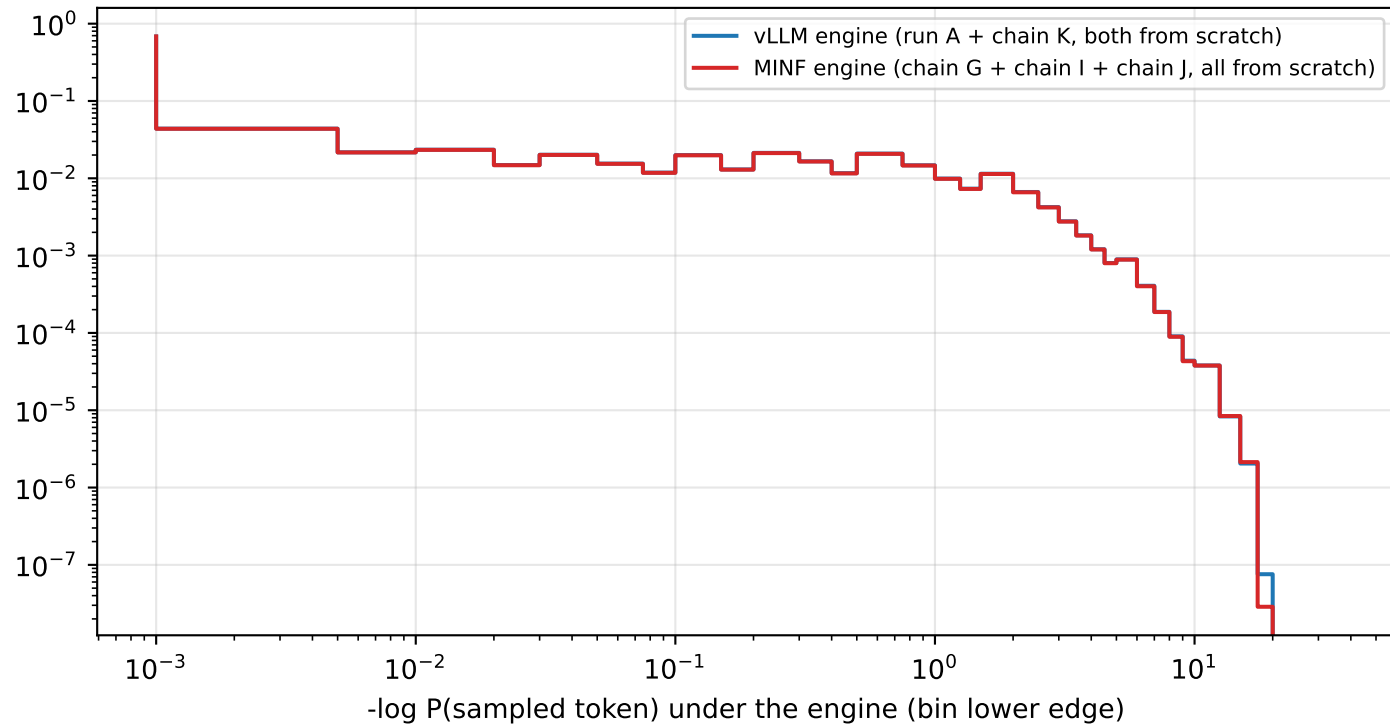


Tail of generated tokens per assistant turn (whole turn incl. tool call)

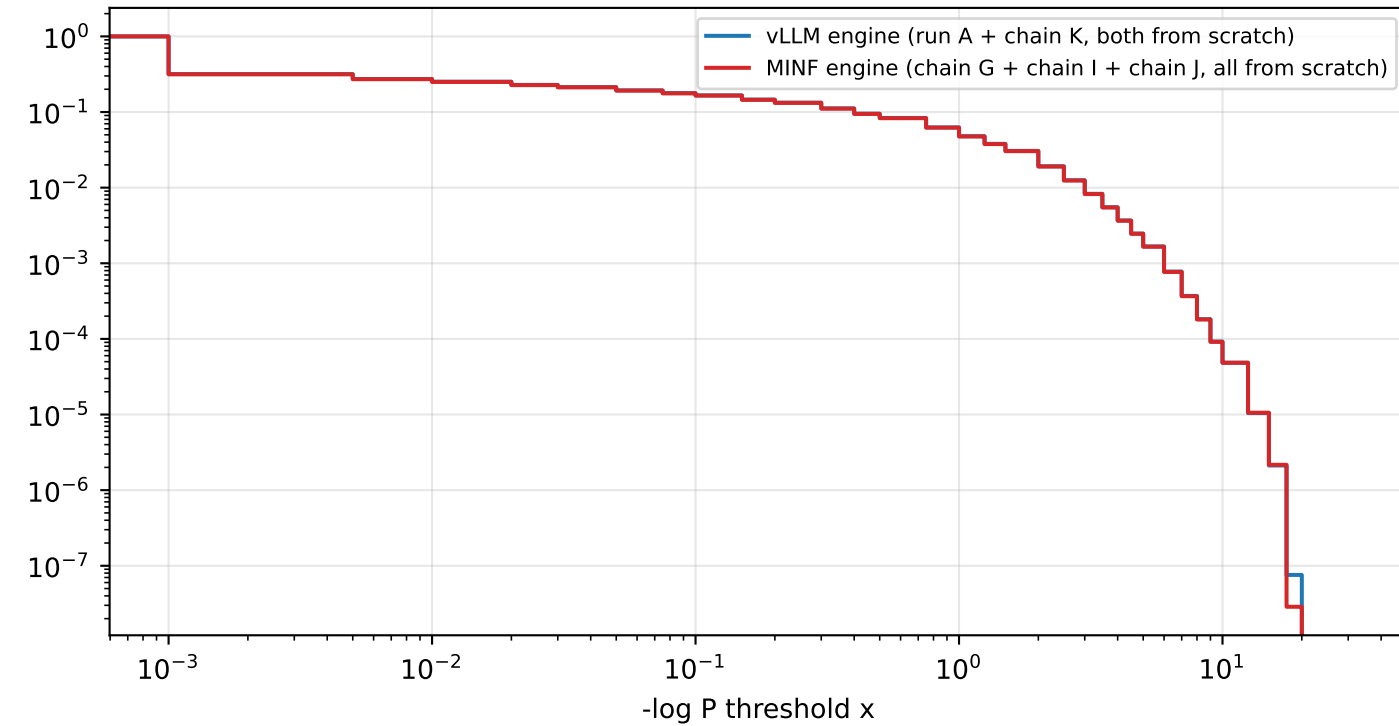


[engine-pooled] Token-level engine logprob distribution including the tail

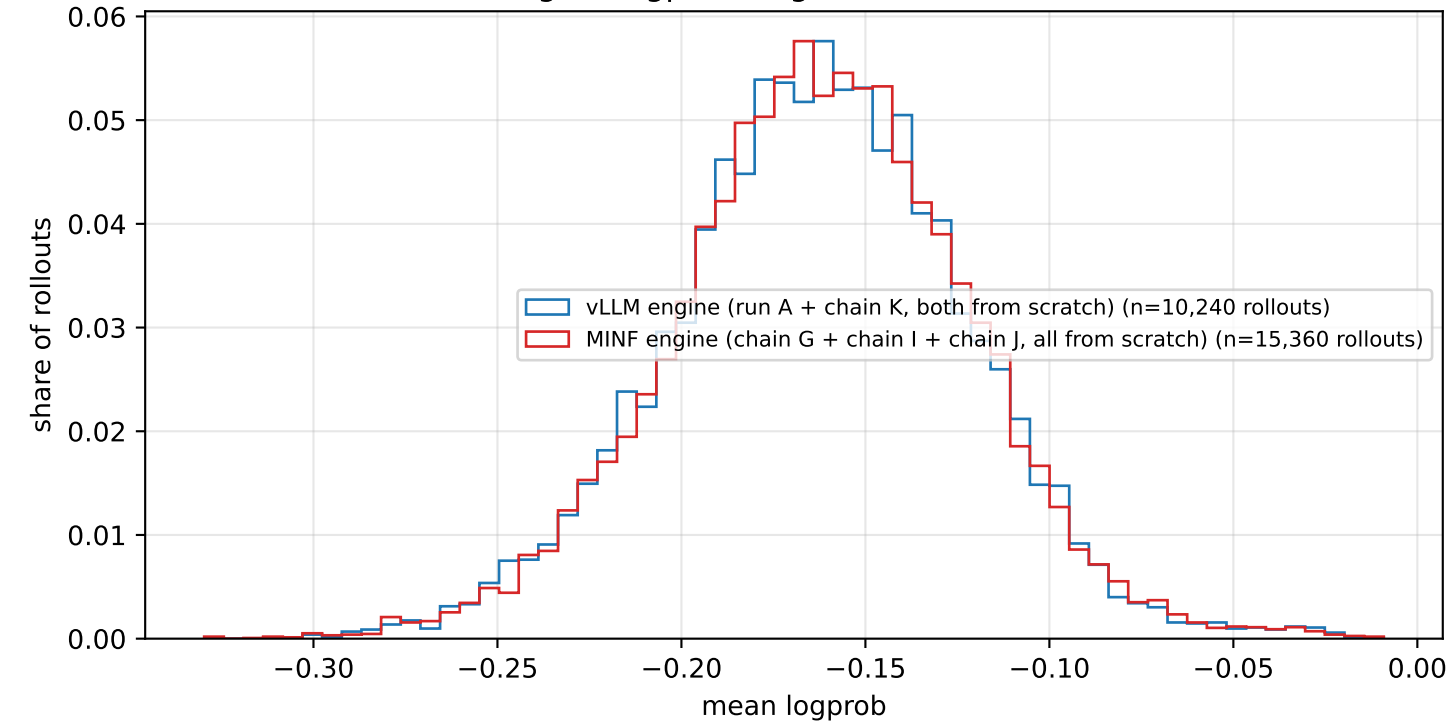
Engine logprob of sampled tokens, share per bin



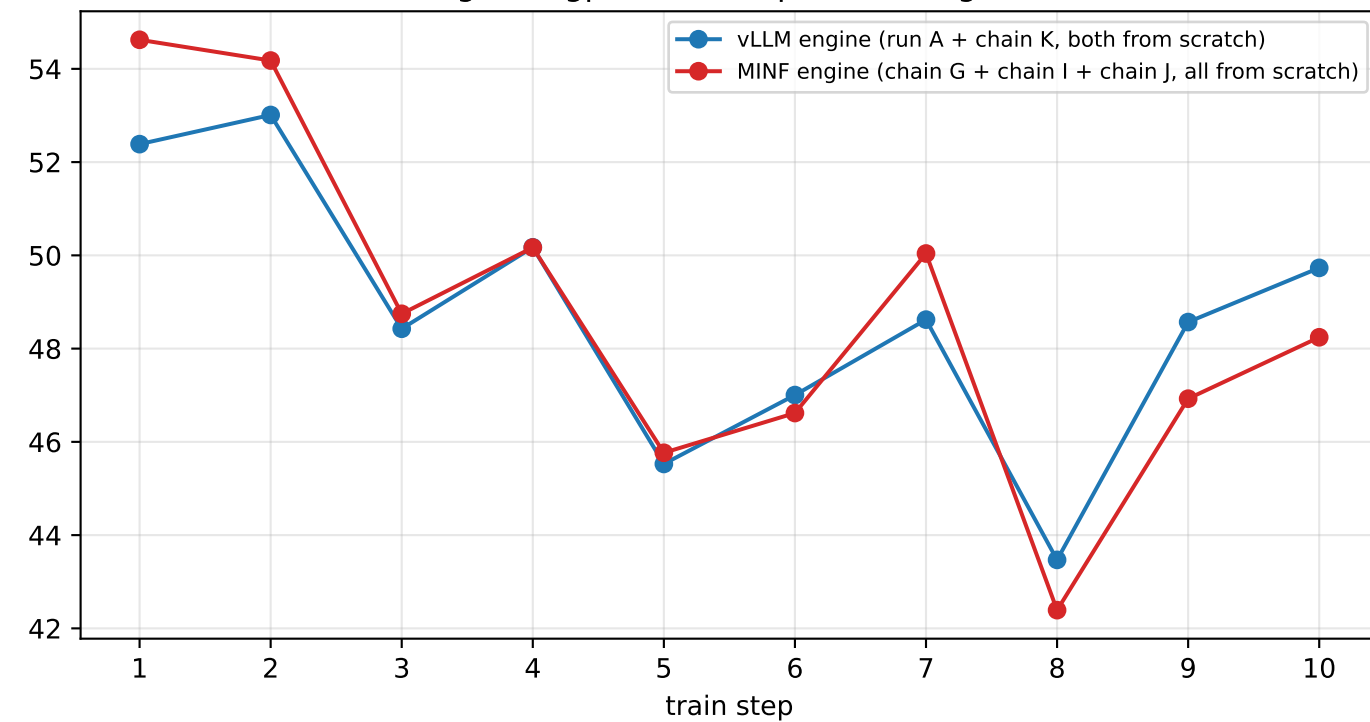
Tail: share of tokens with $-\log P \geq x$



Per-rollout mean engine logprob of generated tokens (share of rollouts)

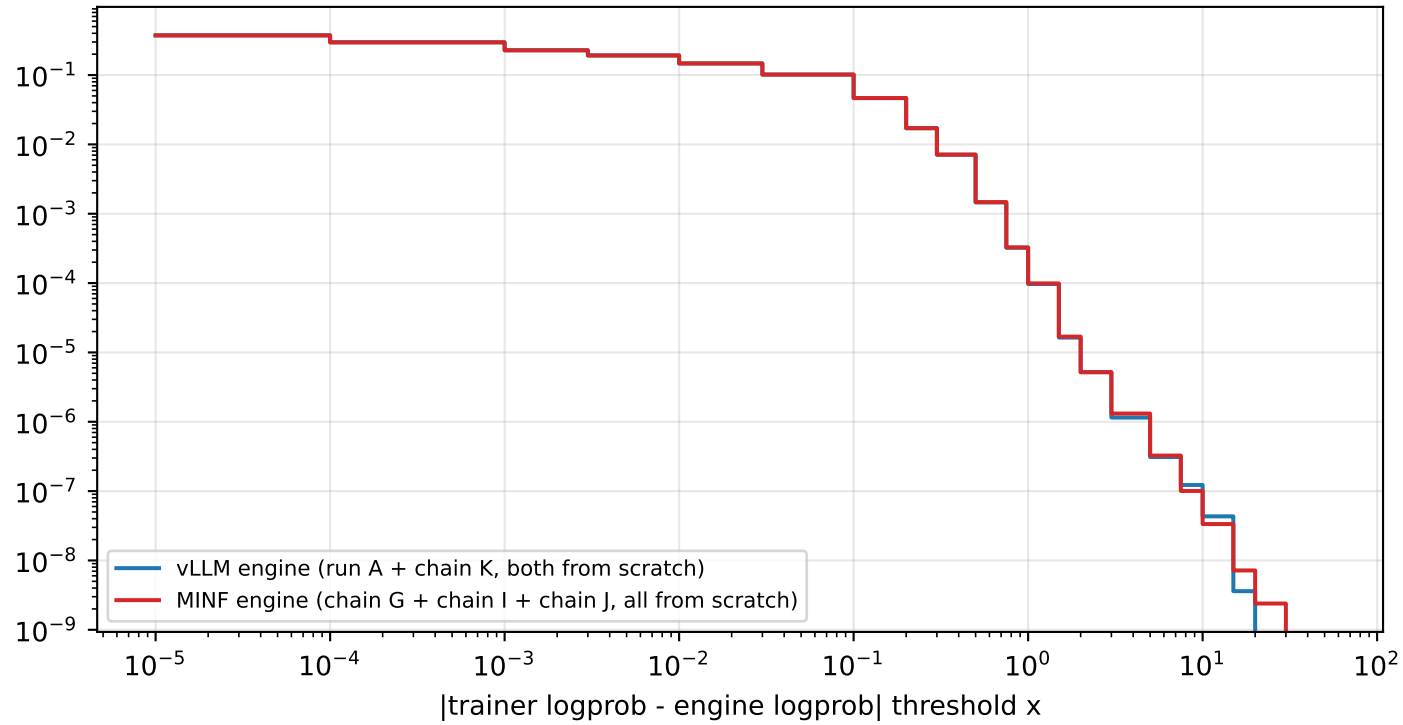


Tokens with engine logprob < -10 , per million generated tokens

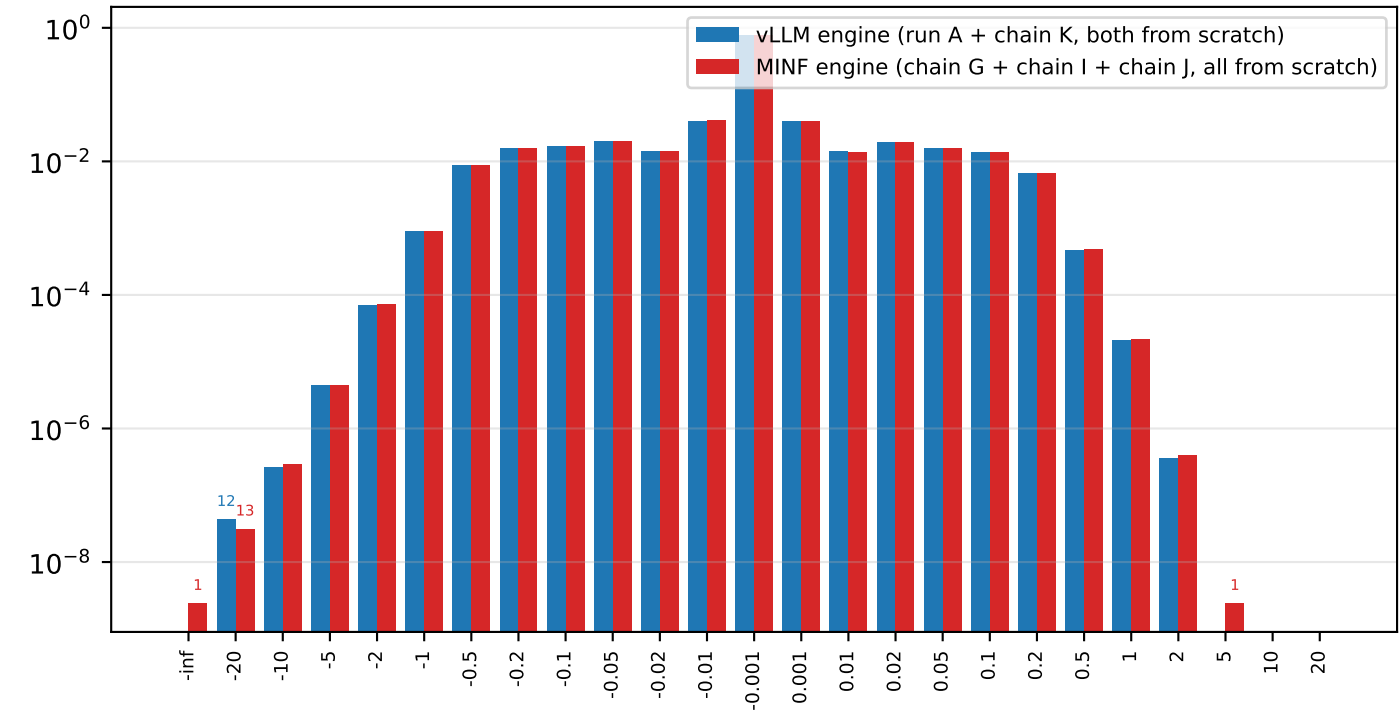


[engine-pooled] Trainer vs engine logprob mismatch

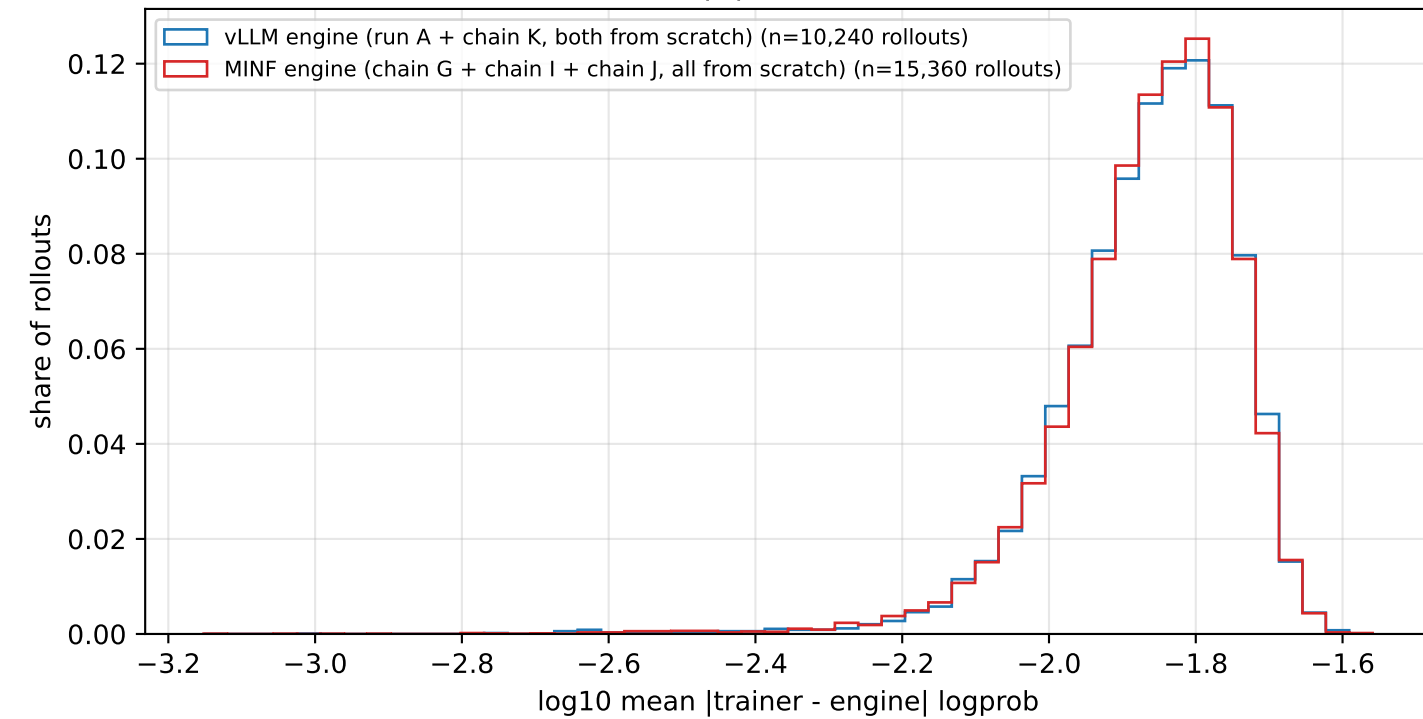
Tail: share of tokens with $|d| > x$



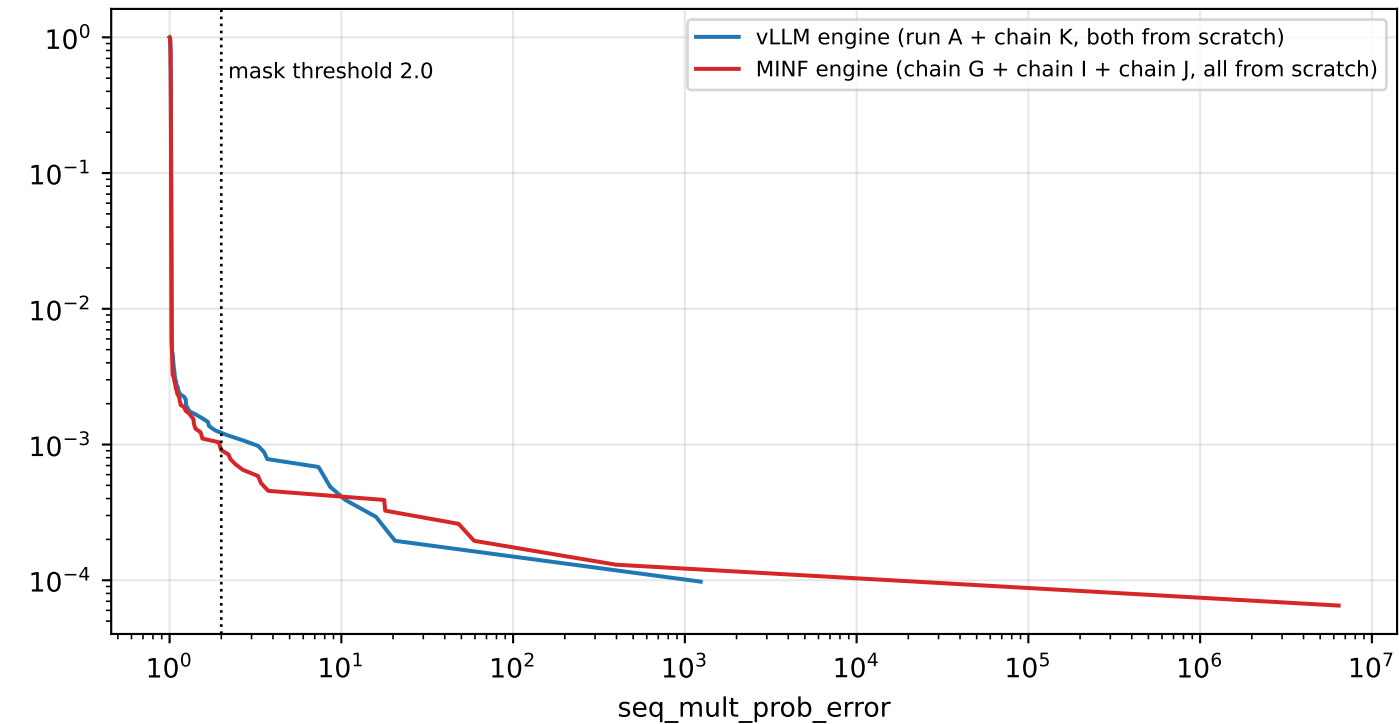
Signed $d = \text{trainer} - \text{engine logprob}$ (bin lower edges; leftmost bin = $d < -20$)



Per-rollout mean $|d|$ (log10), share of rollouts

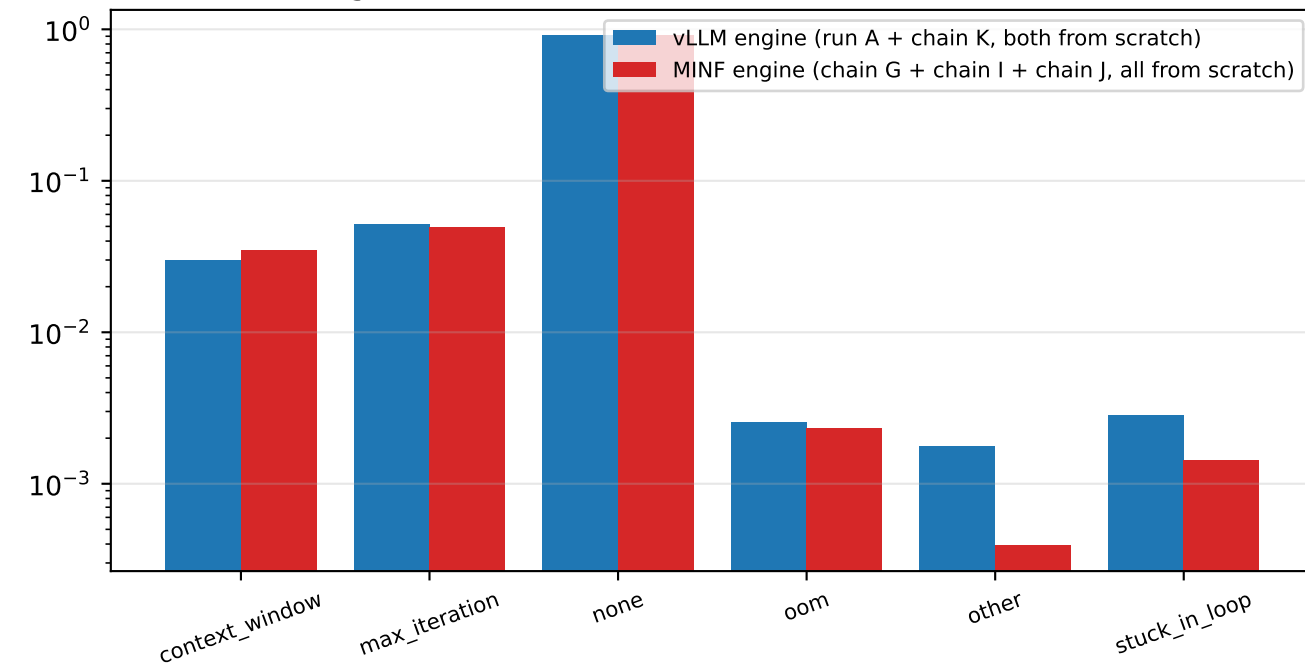


Tail of per-sequence multiplicative prob error

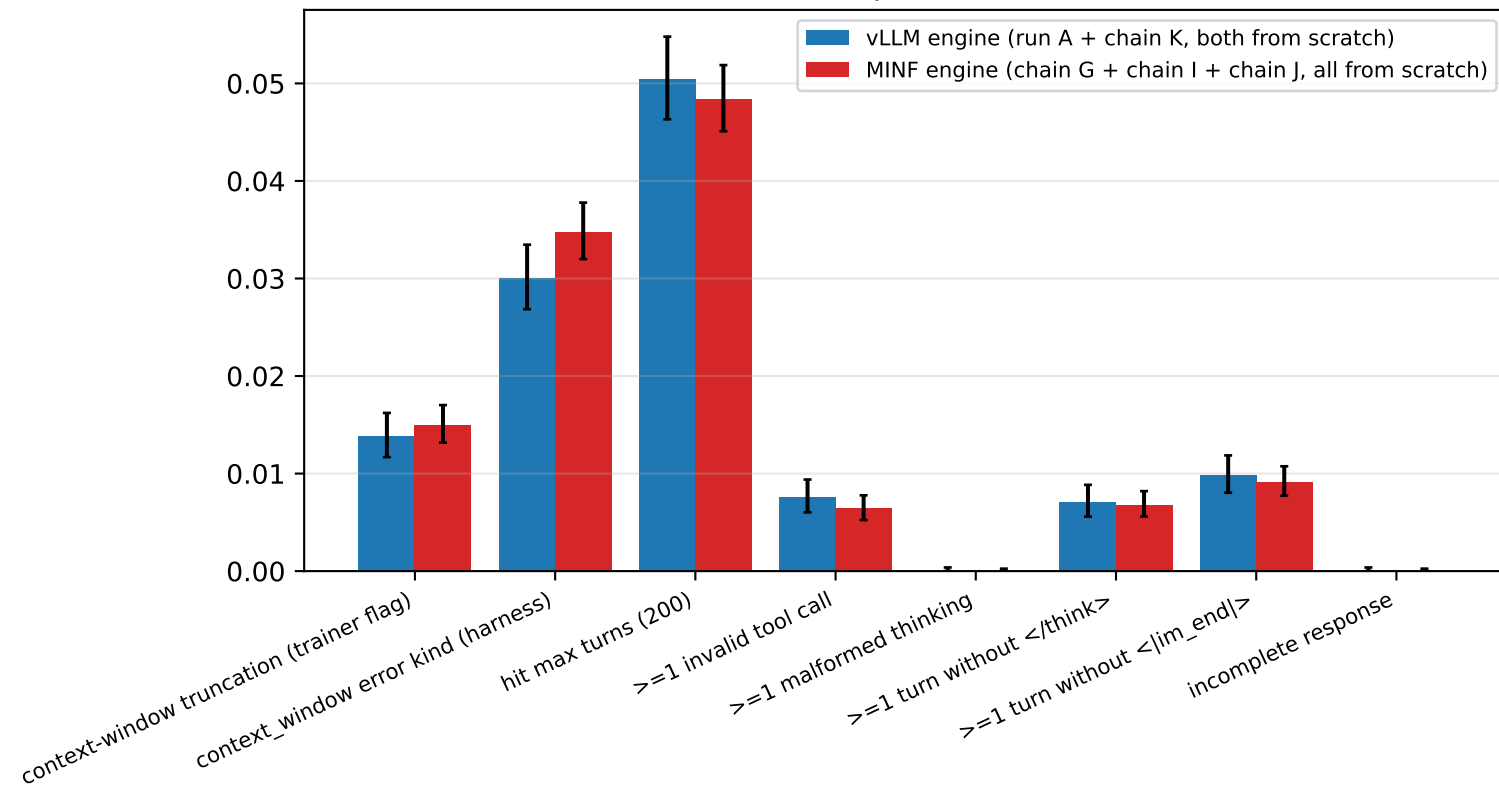


[engine-pooled] Invalid tool calls, malformed thinking, generation limits and termination kinds

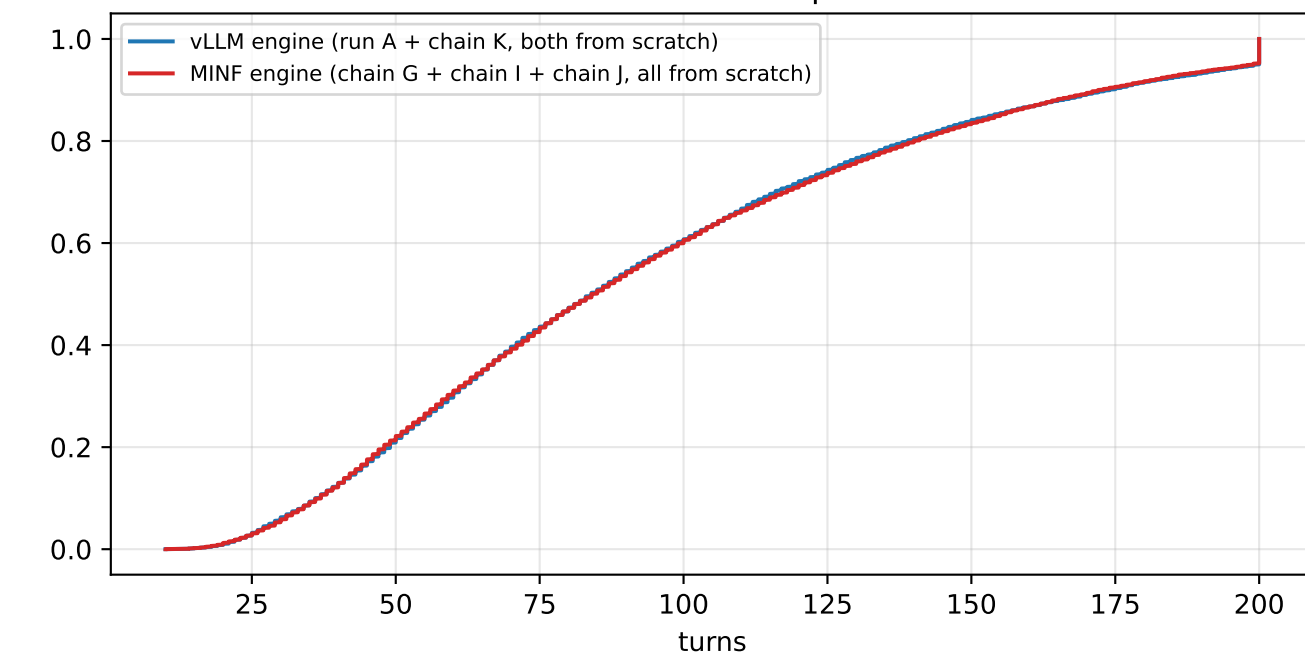
Agent termination / error kind (share of rollouts)



Limit and failure rates per rollout (95% Wilson CI)



CDF of assistant turns per rollout



Per-turn rates, per 1,000 assistant turns

